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Introduction

In January, my cat Mew was diagnosed with diabetes. I noticed the early signs, but panic led me to avoid the issue rather than face it. I tried managing it through diet alone without understanding feline nutrition, which only made things worse.

That was my first lesson: don't delay diagnosis and treatment. Early detection and treatment are key. The longer you delay, the more effort you will need to have to get it managed, and hopefully resolved.

Most cats have Type 2 diabetes and if you catch it fast enough, you can get the cat to a state where you do not need insulin and it can be easily managed by controlling the cat's diet.

I'm sharing my experience in hopes that it encourages others to take action sooner. Early diagnosis and intervention are key. While diabetes can't be cured, it can be managed.

Every cat is unique, and treatment isn't one-size-fits-all. Mew is a 10-year-old Siamese: intelligent, stubborn, and thrives on structure. I wasn't prepared for how resistant he'd be, and that created real challenges.

One of my biggest fears was hypoglycemia, a sudden drop in blood sugar that could be life-threatening. I was terrified I wouldn't catch it in time before needing emergency care. That scenario nearly came true more than a few times, but thankfully I had learned enough to spot it early and act fast.

I'm deeply grateful to the vets who supported us. At the beginning, I didn't even know what questions to ask, but over time, I learned. This document is meant to help you start off better informed when you cannot call the vet and ask the right questions when you have the opportunity.

You won't always have help in the moment, especially at 2 a.m. when your vet is unavailable. You'll need to cope and act confidently until you can seek professional guidance.

I share this in the hope that your journey is shorter, smoother, and less scary. If you have questions, please consult your vet - they're your best resource. This story is meant to support you, not replace expert advice.

What is Type 2 Diabetes?

Type 2 diabetes is a chronic condition where the body struggles to regulate blood sugar (glucose) properly. Chronic high blood glucose is called Hyperglycemia because glucose builds up in the bloodstream. Untreated, it can lead to kidney and liver issues, nerve damage, and eventually death from organ failure or other complications.

Cats with type 2 diabetes struggle to regulate blood glucose (sugar).

- After a cat eats, food is converted into glucose (sugar).
- Glucose is stored in muscles (for energy) and in the liver (to maintain blood sugar levels when needed) excess sugar is handled by the kidneys and expelled in urine.
- Insulin is essential for moving glucose into the muscle tissues.
- The pancreas may not produce enough insulin.
- The body not respond effectively to insulin (called insulin resistance).
- Risk factors are: neutered/spayed, age, gender, weight and breed.
- My cat is a neutered male Siamese, over 10 years old and the resulting risk is very high. All his brothers are quite large.

Symptoms (Detection)

Early symptoms can be subtle and may go unnoticed, especially if your cat roams outdoors. As the condition progresses, symptoms become more obvious. Watch for signs that the body is struggling with excess glucose, such as:

Vomiting

- This was the first sign I noticed and ignored. Cats will also vomit hairballs, originally a reflex from eating hair contributions from furry prey, and less so from hair ingested from self cleaning.
- Vomiting food is not normal, and from my experience can also affect cat eating habits over the long term.
- My cat will not eat like a dog, but will nibble food over time as a conditioned response, probably because overeating due to endless hunger repeatedly causes gastric distress.
- Every person I know with a diabetic cat that is not under control has cat vomit on their floors.
- I wake every time I hear a cat vomit because I know not to ignore it.

Increased thirst and urination

- One of the body's mechanisms to deal with high blood glucose is to use the kidneys to pass the glucose via urination.
- Cats as a species evolved in an arid (dry) region and most of their water comes from the food they eat.
- Because cats should normally get most of their hydration from their food, frequent drinking and urination is not a sign of a healthy cat.
- Even in a multi-cat household, the increased frequency of filling the water bowl or fountain will be noticed.
- This can lead to kidney problems if it is not caught and treated in time.

Appetite and lethargy/ weakness.

- Because the glucose is not being absorbed by the muscle tissues, the cat will always be hungry.
- The more it eats, the higher the blood glucose goes, but the muscle tissue cannot absorb it.
- The less energy in the muscle tissues, the less active the cat will be.
- This is a vicious cycle that can only be broken by administration of insulin.
- The problem is that many senior cats do sleep a lot and so these signs can be easily missed.

Litter Box

- A cat that does not go outside only has one place to do its business - the litter box.
- A cat that is eating and drinking more due to diabetes will show an impact in the litter box.
- Originally, I noticed my little buddy was peeing more often, and I suspected crystals in the urine and put him on what I thought was the right food to correct it.
- The saturation of the litter from urine increased and I started to suspect he had diabetes, but dismissed it.

Coat and grooming

- A sick cat with low energy does not groom a lot. You will always notice health issues by the coat health.
- My little buddy is obsessive about grooming, so much that he will remove any sensor I put on him in very little time.
- He stopped grooming suddenly and I just discounted it as age, a mistake.

Weight loss

- My cat started out heavy, not obese but definitely large.
- His overall weight loss happened over time and it was not really noticed.
- I put it up to his age, I had never known an old Siamese of his body type.
- He definitely lost weight and I was not watching for that.

Neuropathy

- Cats normally walk on their toes (digitigrade stance).
- Neuropathy refers to damage or dysfunction of the nerves, especially those providing feedback and control of the hindquarters, and progressing from there.
- Neuropathy causes weakness, muscle wasting, and loss of coordination in the hind legs.
- In cats, the most common cause is diabetes mellitus, especially when blood sugar levels remain high over time.
- As a result, affected cats adopt a plantigrade stance, placing their full foot—including the hock—on the ground.
- This stance is abnormal for cats and can look like they're crouching or dragging their back legs.
- Visual Example: <https://youtu.be/oHmnFXYHa2g>
- This was a sign I could no longer ignore.

Verification

- I first noticed the increased urination and thought it was a urinary health issue (crystals).
- I tried to solve that with diet, but didn't understand what I do now about a cat's diet. Then I noticed the litter box was being used more often and there was more urine than normal, but he wasn't showing symptoms of crystals in his urine anymore.
- He was vomiting nightly, and I dismissed that as some illness.
- He got less active, started losing weight, stopped grooming normally and I dismissed it with age.
- I put his funny gait to arthritis from old back injuries.

- Then I had to face it, all boxes were checked and I needed to get him (and me) some help. Did he really have diabetes? It sure looked like it.
- Then one night he got into some leftover mac and cheese, ate it from hunger, vomited it all over the place and it was clear he was going to die.
- The emergency vet bill was expected and fair, but still expensive. I recommend that you go see a vet when you first suspect diabetes and don't wait like I did.

Standard blood test

The first test the vet will perform is a standard blood workup. Further testing can prove it really is diabetes and not a more serious condition that can either complicate treatment, or masquerade as diabetes.

Blood Glucose

What it shows: Measures glucose concentration in blood.

Example values:

- Hypoglycemic (Danger, very low blood sugar): under 3 mmol/L (under 54 mg/dL)
- Normal: 3.9–8.3 mmol/L (70.2-149.4 mg/dL)
- Optimal: 6.5 mmol/L (117 mg/dL)
- Normal Peak (After dinner): 8.3 - 16 mmol/L (149.4-288 mg/dL)
- Diabetic (Long term, accompanied by glucose in the urine): greater than 13.9–16.7 mmol/L ()
- Stressed (Short term, no glucose in urine): 6.7–22.2 mmol/L (120.6-399.6 mg/dL)
- Hyperglycemic (Long term, definitely diabetic): greater than 22.2 mmol/L (399.6 mg/dL)

Liver Function

What they are: Chemicals produced by the liver to help digest fats and store the energy.

What it shows: Assesses liver function. A bile acids test checks how well the liver processes and recycles these compounds.

- The liver acts as an energy storage device for excess blood glucose not used by the muscles.
- Chronic hyperglycemia can lead to fat accumulation in the liver (hepatic lipidosis), especially in overweight or insulin-resistant cats.
- Elevated bile acids may signal liver dysfunction, which can occur due to hepatic lipidosis (fatty liver), pancreatitis, or insulin resistance.
- Elevated ALT or ALP might signal that diabetes is affecting other organs, prompting timely intervention.

Example values:

- ALT (Alanine transaminase): Normal < 100 U/L. > 100 U/L: Often a red flag for liver cell damage, it's a key player in the body's metabolism, helping convert proteins into energy for liver cells.
- AST (Aspartate transaminase): Normal < 50 U/L Plays a role in amino acid metabolism. Elevation may suggest bile duct obstruction, steroid use, or liver or muscle injury.
- ALP (Alkaline phosphatase): Normal < 60 U/L Less specific, plays a key role in breaking down proteins and is particularly active in areas of growth or repair, helps confirm liver vs. muscle origin when paired with ALT.

Kidney Function

What they are: These are waste products normally filtered out by the kidneys. If the kidneys aren't working properly, they build up in the blood

What it shows: Assesses kidney function.

- Chronic high blood sugar can damage the kidneys over time (a condition known as diabetic nephropathy in humans).
- Diabetic cats are prone to dehydration, which can elevate BUN and worsen kidney strain.

Example values:

- BUN (Blood Urea Nitrogen): Normal 5 - 13 mmol/L (90-234 mg/dL): - BUN measures the amount of nitrogen in your blood that comes from urea, a waste product formed when your body breaks down protein.
- Creatinine: Normal 53 - 212 µmol/L: - Creatinine is a waste product from the normal breakdown of muscle tissue. It's produced at a fairly constant rate and cleared by the kidneys.

Pancreas Function

What they are: The pancreas plays a dual role—producing digestive enzymes and regulating blood sugar via insulin.

What it shows: This substance is released into the bloodstream when the pancreas is inflamed.

- Inflammation can impair insulin production or worsen insulin resistance.

Example values:

fPLI (Feline Pancreatic Lipase Immunoreactivity):

- Normal: <3.5 µg/L
- Pancreatitis likely: >5.4 µg/L

Ketones

What they are: Ketones are chemical compounds produced when the body breaks down fat for energy instead of using glucose.

What it shows: Indicates diabetic ketoacidosis.

- DKA is a life-threatening emergency where high blood sugar and ketone buildup cause the blood to become acidic.

Example values:

- Normal: Negative
- Diabetic (DKA): Positive

Urine Sample (Glucose)

The vet always tests urine for glucose if diabetes is suspected. This helps rule out elevated blood glucose levels due to stress.

What it is: Detects glucose in urine.

What it shows: This is a sign that blood glucose is not being properly handled and there is excess being processed by the kidneys.

Example values:

- Normal: Negative
- Diabetic: Positive

Fructosamine

This is another test that is always done if diabetes is suspected.

What it is: A compound formed when glucose binds to proteins in the blood (mainly albumin).

What it shows: Reflects average blood glucose over the past 1–3 weeks — unlike a single glucose reading, which can be skewed by stress.

- Elevated fructosamine confirms persistent hyperglycemia, helping distinguish true diabetes from stress-induced glucose spikes.
- It's not affected by short-term stress, so it's a reliable marker even if your cat freaks out at the vet.

Example values:

- Normal: 190–365 µmol/L
- Diabetic:
 - Prolonged hypoglycemia: <300 µmol/L
 - Excellent control: 350–400 µmol/L
 - Good control: 400–450 µmol/L
 - Fair control: 450–500 µmol/L
 - Poor control: >500 µmol/L

Other medical conditions to consider

A cat diagnosed with diabetes may be difficult to control in terms of blood glucose levels. Other factors may be impacting the cat's immune systems such as infections, steroid use. Here are some to consider.

Acromegaly (also called hypersomatotropism)

A vet would typically recommend testing a diabetic cat for acromegaly when the cat shows signs of insulin resistance, meaning the diabetes is difficult to control despite high doses of insulin.

What it is: Caused by a growth hormone-secreting tumor in the pituitary gland leading to excessive growth hormone (GH) and elevated insulin-like growth factor 1 (IGF-1). It most commonly affects older male cats (8–14 years old).

What it shows: Elevated IGF-1 confirms excess Growth Hormone activity

- Usually treated with radiation therapy or surgery.
- Untreated acromegaly leads to progressive insulin resistance, making diabetes nearly impossible to control.
- Cats may develop organ enlargement, arthritis, and neurological symptoms.

Symptoms:

- Insulin-resistant diabetes (high insulin doses with poor glucose control)
- Weight gain despite diabetes
- Broad facial features, enlarged paws, jaw, or tongue
- Organ enlargement (heart, liver, kidneys)
- Stridor or breathing changes due to soft tissue growth

Example values:

IGF-1

- Normal: <100–300 ng/mL
- Acromegaly: >1000 ng/mL (often >1500 ng/mL)

Thyroid

A vet would recommend testing a diabetic cat for thyroid issues, especially hyperthyroidism when certain signs or complications arise that suggest the thyroid may be interfering with diabetes management.

What it is: The thyroid gland, produces hormones that regulate metabolism, affecting: heart rate, body temperature, digestion, energy levels and weight control.

What it shows: Helps rule out or confirm thyroid issues that may be complicating diabetes treatment.

Symptoms:

- Hyperthyroidism (overactive thyroid) can mimic or mask diabetes symptoms, such as:
 - Weight loss despite good appetite
 - Increased thirst and urination
- It can also interfere with insulin regulation, making diabetes harder to manage.

Example Values:

- Total T4 (Thyroxine): Normal: 1.0 – 4.0 µg/dL
- Free T4: Should align with total T4 if thyroid is normal
- TSH (Thyroid Stimulating Hormone): < 0.03 ng/mL

Fanconi syndrome

If a cat has reasonable blood glucose levels, with glucose in the urine, Fanconi syndrome should be considered.

What it is: Fanconi syndrome in cats is a rare disorder of the kidneys, specifically affecting the proximal renal tubules, which are responsible for reabsorbing vital substances from the urine back into the bloodstream.

Symptoms:

- Muscle wasting
- Poor body condition
- Acidosis (a condition where the body's fluids become too acidic)
- Increased urination and thirst (though not always present)

Gingivitis

Gingivitis occurs when plaque, a sticky film of bacteria, accumulates on the teeth and irritates the gums. If untreated, it can progress to periodontitis, a more severe and irreversible condition that damages the structures supporting the teeth.

Why it is important to control: Diabetic cats are more vulnerable to gingivitis due to their compromised immune systems and elevated blood sugar levels:

- Higher susceptibility to infections: Diabetes weakens the immune response, making it harder to fight off gum infections.
- Poor blood sugar control: Infections like gingivitis can worsen diabetes by making it harder for the body to regulate glucose levels.
- Vicious cycle: High blood sugar promotes gum disease, and gum disease makes blood sugar instability worse.
- Loss of appetite: Pain from gingivitis may cause diabetic cats to eat less, risking dangerous drops in blood sugar or diabetic ketoacidosis.

Symptoms:

- Red, swollen gums (especially along the gumline)
- Bleeding when gums are touched or teeth are brushed
- Bad breath (halitosis)
- Drooling
- Difficulty eating or chewing (may favor one side of the mouth)
- Crying or yowling during meals
- Weight loss
- unkempt fur due to pain while grooming

Treatment

Diet

Diet is one of the most important factors for cat health. Unfortunately, we mostly trust the claims of the cat food producers' claims with very little knowledge of why they may be false. Shopping for cat food, we often look for foods our cats prefer, and this can lead to disaster over the long term.

- Cats are obligate carnivores, meaning they must eat meat to survive. This isn't just a preference, it's a biological necessity shaped by their digestive system, metabolism, and nutritional needs.
- They have a relatively short gastrointestinal tract, optimized for digesting animal protein and fat, not plant matter.
- Their stomach acid is strong, helping break down raw meat and kill bacteria.
- They produce very little of the enzyme amylase, which breaks down carbohydrates. This means carbs are poorly digested, leading to rapid glucose spikes in the bloodstream.
- Cats require nutrients that are only found in animal tissue or are poorly available in plants:

Needed	For	Found In
Taurine	heart, eyes, and reproduction	Meat only
Arginine	detoxifying ammonia	Meat only
Vitamin A (Retinol)	Cats can't convert beta-carotene	Liver, animal fat
Niacin (B3)	Cats can't synthesize it from tryptophan	Meat
Arachidonic acid	Essential fatty acid	Animal fat
Vitamin D (D3)	Must come from diet—not sunlight	Fish, liver

- Diets lacking these nutrients can lead to:
 - Blindness (from taurine deficiency)
 - Liver and kidney damage
 - Poor coat quality
 - Neurological issues

Food composition and digestion

- The ideal food for a cat should contain a high percentage of animal protein, a moderate amount of fat, and a low percentage of carbohydrates.
- Unfortunately, most cat foods do not have the necessary labelling to properly indicate the true levels of these components and it can be nearly impossible to properly determine the best food for your little furry buddy.

Carbohydrates

Carbohydrates will have the earliest and most dramatic effect on blood glucose. As mentioned before, they are poorly digested and will cause spikes in blood glucose levels.

- Sugars are faster acting and have more dramatic effect than starches which require more digestion.
- Carbohydrates are not totally bad, they do have their use, especially countering the rapid drop in blood glucose levels after an insulin injection.
- All wet cat foods with gravy are over 10% carbohydrates. Most pates are 5% or less.
- A food with less than 5% carbohydrates is suitable for diabetic cats, or cats at risk.
- Most prescription foods (there are only two) are strictly controlled to have consistent carbohydrate levels under 5%.
- Since non-prescription food labels do not accurately reflect the composition, it is necessary to use prescription foods for diabetic cats, or use some simple rules for cats you wish to control to avoid diabetes.

Fats

- Fats will generally show up in blood glucose after carbs. This will become more evident with proper blood glucose monitoring.
- A lot of cat food manufacturers will use fats over more costly proteins in their foods. As a rule, choose a food with less fat content than protein.

Protein

- Protein does not have as dramatic and immediate effect on blood glucose levels, it will have a moderate effect over time, but will not cause spikes in blood glucose levels.
- Because of the importance of the nutrients in meat-based protein for cat health, be wary of vegetable-based proteins in cat food.

Wet Versus dry food (content, hydration and digestion)

All future discussions refer to pate formulations when referring to wet food.

- Cats evolved as desert animals and have a low thirst drive. Wet food contains 70–80% moisture, helping prevent:
 - Dehydration
 - Urinary tract issues
 - Kidney disease
- Dry food has only 10% moisture, which can strain the kidneys over time.
- Dry food often contains fillers like corn, wheat, or rice (carbohydrates), which cats struggle to digest.
- Wet food is typically higher in animal protein and lower in carbohydrates. This supports:
 - Lean muscle maintenance
 - Stable blood sugar levels
 - Reduced risk of obesity and diabetes
- Wet food mimics the moisture and nutrient profile of prey animals. It provides:
 - Taurine, arginine, and vitamin A, essential nutrients found in meat
 - A texture and aroma that's more palatable, especially for picky or older cats

Pate versus gravy

If you shop in most grocery stores for cat food, you will find a larger variety of gravy type wet foods than pate. People buy what cats love and eat readily, but they don't understand the dangers in this decision.

- Gravy requires a starch (carbohydrates) as a thickener, so logically, gravy-based foods typically have 3-4 times more carbohydrates than pate foods.
- It is not uncommon for a gravy-based food to be between 12-14% carbohydrates. They still have a use, but not for regular feeding.
- Most pate wet foods have less than 5% carbohydrates, the threshold for what you should regularly feed a diabetic cat.
- Most fish-based pates have 1-2% carbohydrates, poultry-based pates have between 2-4%.
- These numbers can change between batches; quality control is not high in these foods and component percentages can change batch by batch.
- The pate you can find on most grocery stores - Fancy Feast Tender Chicken and Liver, or Turkey and Giblets are about 2% carbohydrates from the information I have.
- I can buy Purina Fancy Feast pate (3oz can) from a local grocery store, but can get the same from Amazon for about 30% less delivered to my door.
- If you want a good pate with no carbs, I suggest the Tiki Kat "after Dark" soft pate cans or the velvet mousse pouches. It mixes well with my cat's diabetes food and enhances the flavor with few immediate and unpredictable effects on blood glucose levels.

Poultry, rabbit, etc. versus fish

A best practice for any animal is to feed it what it would normally obtain in the wild. Any cat owner will appreciate the hunting skills of their outdoor cats.

- The primary targets are small mammals (mice, rabbits) and birds.
- Unless your cat is overly large, sheep, goats and cows are safe, and given cats don't normally like water, fish is also off the list.
- This does not mean that if there is an easily available carcass, the cat will avoid it, but it's not normally on the menu.
- You are not going to find much variety in cat food protein sources, the most common will be poultry (chicken, duck, quail or turkey) or fish.

Feline tooth resorption (Vitamin D, fish oils)

Overconsumption of certain fish oils in cat food may contribute to feline tooth resorption (also known as Feline Odontoclastic Resorptive Lesions (FORL)) through its impact on vitamin D metabolism and immune regulation, although the exact mechanism is still under investigation.

- This is a painful condition where odontoclasts (cells that break down tooth structure) become overactive.
- It leads to progressive destruction of the cementum, dentin, and enamel, often starting below the gumline.
- Fish oils can be rich in vitamin D, especially vitamin D3 (cholecalciferol). Excess vitamin D may:
 - Alter calcium and phosphorus balance
 - Stimulate odontoclastic activity
 - Promote mineralization abnormalities in dental tissues
- Omega-3 fatty acids in fish oil can modulate immune responses. While generally anti-inflammatory, excessive intake may dysregulate immune cells, including those involved in bone remodeling. This could potentially trigger or accelerate odontoclastic resorption in predisposed cats.
- Diets overly reliant on fish or fish oils may lack other essential nutrients or create imbalanced mineral ratios. This imbalance may contribute to periodontal inflammation, another suspected trigger for resorption lesions.
- While fish oils have benefits for skin, coat, and joints, moderation is key, especially for dental health. If your cat is showing signs like drooling, dropping food, or chewing on one side, it's worth discussing tooth resorption with your vet.

Prescription versus off the shelf

There are both wet and dry cat foods for diabetic cats available from Purina and Royal Canin by prescription.

- The advantage to using the prescription Diabetic food is the predictability of the formulation, no matter how I tried, I just could not get consistent results from off the shelf pate.

- The wet foods differ, Purina's is a pate, (5.5 oz can) and a chunk and gravy ProPlan DM Savory Selects (5.5 oz can) and Royal Canin just recently changed their chunk and gravy (3oz) can recipe. I switched him from Royal Canin to Purina Savory Selects. He's eating a lot more, and my only problem is keeping his sister from his food.
- My little buddy would rather starve than eat something he doesn't like. It's important that when I feed him, he eats as much as he can and as soon as he can, so my choice was made for me. He would not eat pate food, and only would eat the Royal Canin if I mixed it with a pate with flavor, so I am grateful that I do have an option.
- I was advised that the gastro Royal Canin food could substitute for the diabetic, but found my cat disliked it. Apparently, this dislike is true for all gastro foods.
- The disadvantage to the Diabetic food is that it only comes in one flavor. Again, being the picky eater he is, he does like variety.
- My solution to this is to take the Royal Canin "gravy", mix with:
 - 1/4 tsp of salmon oil or sunflower oil (or not) or Fortiflora with a little warm tap water (or not)
 - 1/2 oz of Tiki Kat pate or mousse to make a "dressing"
 - Mix it with the rest of the Royal Canin food.
 - This gives him some variety which he will accept.
- The vet has an online store I can get the prescription foods from and have delivered in a couple of days to my door, a great convenience.
- I tried to calculate the Royal Canin carbohydrate content as best I could, coming out to about 4.6%. It's not an exact science and you will not get accurate information at all from the manufacturer.

Get a structure for feeding and injections

If you want predictable results, you need consistent inputs. There are so many variables you need to account for. The more variables that you can control, the less variation you will see.

- When I started, and the vet could not answer why I was having such good results in the morning, but horrible results in the evening. The root cause is that I would feed him later in the morning (5:30), inject him quickly so I could be at work by 6:00am, and then I could take my time in the evening. By changing the times, I should have expected different results.
- I have my cat injection timed for 5:30am and 5:30pm, exactly 12 hours apart. He is fed at exactly 5:00 and will usually take 30 minutes to eat enough food to allow me to inject him. Now that I start exactly 12 hours apart, giving him 30 minutes to eat and then give him his injection, the results are much more consistent.
- This gives me time to gauge his appetite and give his meal time to start to take effect. If he has eaten about 50% of his food, his appetite works and I can safely give him his injection. If not, I may have to wait, or reduce his dosage based on my estimate of how much he will eat. This judgement comes with time and experience. Mistakes can result in very undesirable results.

Time from feeding and insulin effect

For diabetic cats, feeding low-carb wet food and timing insulin properly is crucial to maintaining stable glucose levels.

Insulin

When insulin is injected subcutaneously in cats, the onset of absorption and action depends on the type of insulin used.

- Factors like hydration, body temperature, and injection site can affect absorption speed.
- I inject my cat in the shoulder or neck mostly because he is wearing a shirt and this is the most accessible location that does not interfere with, or move his Glucose sensor. As well, he rarely resists the injection as opposed to other locations.
- I do see gentle effects within about 30 minutes, but the stronger effect usually takes an hour or so.

Wet food

When a cat eats wet food, the impact on blood glucose levels typically begins within 30 minutes to 1 hour after ingestion.

- Timeline
 - 0–30 minutes: Digestion begins in the stomach; carbohydrates start breaking down.
 - 30–60 minutes: Glucose from digested carbs enters the bloodstream, causing a rise in blood sugar.
 - 1–2 hours: Peak glucose levels are usually reached, depending on the food's composition.
 - 1–2 hours: Blood glucose begins to decline as insulin facilitates glucose uptake.

- If you are feeding diabetic wet food, or a high carb wet food to counter a potentially dangerous low blood glucose situation, be aware that the effect is not immediate, it can take a few minutes to start affecting the blood glucose level and do not panic. It is best to set a limit, 5.5 mmol/L or less as a trigger to feed the cat so it has time to take effect.
- Foods with simple sugars or starches spike glucose faster. Diabetic wet food is designed to work with insulin, so will take effect faster and assist countering the effect of insulin faster. I usually see effects within 5-15 minutes after feeding.
- I have found my cat will eat just enough and usually in enough time to deal with it. I can't explain how he knows this, but it usually is the case if I catch it fast enough.

Dry food (Diabetic Prescription)

When a cat eats dry diabetic prescription food, the impact on blood glucose levels typically begins within 30 minutes to 1 hour, but the magnitude and duration of the glucose rise are generally less pronounced because these foods are formulated to minimize post-meal glucose spikes.

Hunger and Grazing (why you would allow it)

Cats experience hunger due to a combination of physiological, behavioral, and environmental factors.

Physiological Hunger Triggers

- **Empty stomach:** Like humans, cats feel hunger when their stomach is empty and ghrelin (the "hunger hormone") is released.
- **Blood sugar dips:** Low glucose levels can prompt the brain to signal hunger.

Behavioral and Environmental Hunger Triggers

- **Routine and habit:** Cats are creatures of habit. If they're used to eating at certain times, they may show hunger cues even if they're not physiologically hungry.
- **Food cues:** The sound of a can opening or the smell of food can trigger anticipatory hunger.
- Some form of control over feeding is good to ensure the cat is hungry and eats at the scheduled time and consumes an acceptable amount of food.
- Normally, I will put the dry food away a couple hours before and during feeding times to ensure that the wet food portion is eaten, given that is faster acting and digested faster.
- When you do not have the cat under observation (asleep, at work, etc.), the cat may not have a food source available if blood sugar levels drop unexpectedly.
- In those times, I ensure that the cat has diabetic dry food available. It may not have as fast or dramatic an effect, but it can still help if the cat's hunger triggers.
- My cat normally is full after a 3 oz can of cat food is consumed. In times when his blood glucose dipped dangerously low, he has eaten up to 2 additional 3 oz cans of wet food, despite being full.
- If he has only eaten about 1/2 his wet food and gets his injection, he will normally eat the rest of it when his blood glucose starts to drop under 19 mmol/L.

Medications that affect appetite

As a result of my negligence, by the time the cat went to the vet, he was very ill. He vomited lots, had nausea, was dehydrated and had no appetite. It was imperative to get him to eat his wet food for rehydration and get him stable enough to get his injections.

- The option of pill or orally administered (via syringe) medication was critical since he will fight and spit out pills. I have had many cats that would take medications in pill form, but this guy is stubborn.
- I was given a compounded Mirtazapine in a syringe. A note of caution, do not try to inject it down the throat, aim for the side of the mouth. If the liquid enters the windpipe (trachea) instead of the esophagus, it can be inhaled into the lungs and may lead to aspiration pneumonia, a serious and potentially life-threatening condition.
- After 15 minutes from administration, he was ravenous and ate a can and a half of wet food. It probably contributed a lot to saving his life.
- There are a few options for medication, and I always keep some on hand, just in case.
- Each has different mechanisms and side effect profiles, so your vet will choose the best option based on your cat's health status and preferences.
 - **Elura (capromorelin oral solution):** It's FDA-approved specifically for cats and works by mimicking ghrelin, the hunger hormone, to stimulate appetite and food intake.
 - **Mirataz (topical gel):** Use gloves when applying

- **Cyproheptadine (oral tablet or compounded liquid)**
- **Mirtazapine (oral tablet or compounded liquid):** Tablets can take effect about an hour after administration.

Supplements

Salmon/Fish Oil

Salmon and pollock oil are concentrated dietary sources of omega-3 long-chain fatty acids, primarily EPA and DHA, used as supplements to support multiple aspects of feline health.

Salmon oil can offer anti-inflammatory and supportive benefits that may help some diabetic cats, but because it adds concentrated fat and calories, and because pancreatitis and weight management are important concerns in diabetic patients, it must be used only with veterinary guidance, careful dosing, and close monitoring.

FortiFlora (Probiotic)

FortiFlora is a powdered feline probiotic (single-serve packets) containing live beneficial bacteria formulated to support intestinal microflora, improve stool quality, and support the immune system. It's widely recommended by veterinarians for digestive support, diarrheal episodes, and when changing diets or giving antibiotics.

FortiFlora can be useful for diabetic cats in stabilizing digestion and supporting gut and immune health, which helps keep feeding and insulin routines predictable. It is not a diabetes cure and should be used under veterinary guidance as part of a comprehensive diabetes management plan.

Cat nutrition sources and issues with data

<https://catinfo.org/> ([Feeding Your Cat: Know the Basics of Feline Nutrition, Lisa A. Pierson, DVM](#))

Created by veterinarian Dr. Lisa Pierson, this site features a comprehensive chart comparing protein, fat, carbohydrate, and phosphorus content across various canned cat foods. Ideal for cats with specific health needs like kidney disease or diabetes.

Especially useful: <https://catinfo.org/docs/CatFoodProteinFatCarbPhosphorusChart.pdf>

- This data is somewhat dated and it is not complete by any means, but is still useful. It has fairly good data for Purina Fancy Feast wet food which is readily available at local grocery stores and Amazon.
- Cat food manufacturers are reluctant to provide updated information for their products. I have to give all credit to Lisa Pierson for her persistence. My attempts to get Purina to provide data for two of their newer products have yielded no results after months of trying.

Exercise (using stored glucose)

Exercise plays a vital role in managing feline diabetes, much like it does in humans. It helps regulate blood glucose levels, supports weight management, and improves overall well-being.

- Play is the best form of exercise, having cat trees to encourage climbing and jumping can also help. Use wand toys, laser pointers, or feather teasers to mimic prey and trigger hunting instincts.
- Keep sessions short (5–10 minutes) but frequent—2–3 times a day is ideal.
- Avoid overexertion, especially right after insulin injections or meals, this can cause hypoglycemia.

Insulin

Managing feline diabetes typically involves daily insulin injections, and veterinarians choose from several types based on the cat's individual needs.

Types

Insulin Type	Onset of Action	Duration	Why It's Used
Vetsulin (porcine lente)	~1–2 hours	8–10 hours	FDA-approved; useful for post-meal glucose spikes
ProZinc (PZI)	~1–2 hours	13–24 hours	FDA-approved for cats; good duration and control
Glargine (Lantus)	~1–2 hours	12–24 hours	High remission rates; stable glucose control

Detemir (Levemir)	~1–2 hours	12–24 hours	Potent; used in difficult-to-regulate cases
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Why Vets Choose One Over Another

- **Glucose curve stability:** Long-acting insulins like Glargine and PZI provide smoother glucose control.
- **Remission potential:** Glargine has shown high success rates in achieving diabetic remission in cats.
- **FDA approval:** ProZinc and Vetsulin are specifically approved for feline use, which can influence vet preference.
- **Cat's health status:** Cats with insulin resistance or concurrent illnesses may respond better to certain types.
- **Owner lifestyle:** Frequency of injections and ease of administration matter, some insulins require twice-daily dosing.

Concentration

When prescribing insulin for diabetic cats, veterinarians typically choose from a few well-established types, each with specific concentrations and dosing guidelines.

- Typically, the 100 units/mL (U-100) are human targeted and the 40 units/mL (U-40) are animal-specific.
- U-40 insulin (like ProZinc and Vetsulin) requires U-40 syringes, while U-100 insulin (like Lantus and Levemir) requires U-100 syringes.
- Using the wrong syringe can lead to dangerous dosing errors.

Insulin Type	Concentration	Starting Dose
Vetsulin (porcine lente)	40 units/mL	0.25–0.5 units/kg every 12 hrs
ProZinc (PZI)	40 units/mL	0.25–0.5 units/kg every 12 hrs
Glargine (Lantus)	100 units/mL	0.25–0.5 units/kg every 12 hrs
Detemir (Levemir)	100 units/mL	Often lower than others due to potency (e.g., 0.1–0.2 units/kg)

How to determine the best dosage of insulin to give

Dosing may vary depending on the cat's weight, glucose levels, diet, and concurrent health issues. Work with your vet for best results.

- Most vets begin with 0.25–0.5 units per kg of body weight, given twice daily, then adjust based on glucose curves and clinical response.
- Always follow your vet's instructions closely, and never adjust insulin doses without professional guidance.
- As time goes on, you will become more comfortable with interpreting the results and will gain an understanding of how the insulin dose affects your cat.
- There is a lot to learn, your vet is there to help you. You just need time to absorb it all and get some experience and a better understanding.
- There is a saying - "every cat is different". This document is about my cat, what works for him may not work the same for your cat.

Establishing a curve

A blood glucose curve (BGC) is a diagnostic chart used to evaluate how effectively insulin is managing a diabetic cat's blood sugar levels. It involves measuring blood glucose at regular intervals (typically every 2 hours) over a 12 to 24-hour period following insulin administration.

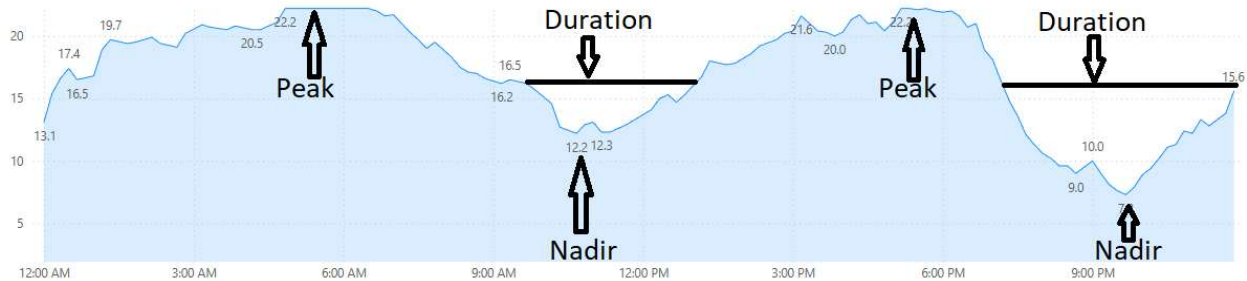
Key Terms

- **Peak (Maximum Glucose Level):** The highest blood glucose reading during the curve.
 - Often occurs just before insulin begins to act or after its effect wears off.
 - Example: ~19 mmol/L
- **Nadir (Minimum Glucose Level):** The lowest blood glucose reading, indicating the point of maximum insulin effect.
 - Ideal nadir range: 4.5–8.3 mmol/L
 - Example: ~6.5 mmol/L
- **Duration of Action:** The time span during which insulin keeps glucose within the target range.
 - Target range: 4.5–16.5 mmol/L

- Example: From hour 1 to hour 10 post-injection

Considerations

The more often you sample the blood glucose, the easier it will be to identify when the nadir occurs, so in the beginning, you may wish to perform sampling more often.



- Every cat is different and you will probably be starting with a less-than-ideal cat. The example curve will not be what you see at the start.
- The vet will offer to perform this initial task, or show you how to do what you need to do at home. It will take more than a few days to properly establish the curve.
- You need to consider that the cat may have higher glucose levels at the vet due to stress. It is really important that to cat is comfortable and relaxed, and doing this at home will probably have better results.
- There are some advantages to using a Continuous Glucose Monitor (CGM), most importantly that it logs glucose readings more often, and they can be shared with the vet over the internet. Be sure to ask your vet about this. There is more information for using a CGM later in this document.
- If you choose to use the ear-prick method, you will need to be sure to maintain a log of insulin dose and time given, timing of meals and time and blood glucose levels observed.
- Just like every cat is different, every day is different as well. No two curves will be the same, some will be disappointing, some will be not so disappointing. The goal is to determine over time what insulin dose works best.
- In my case, I inject my cat at 5:30am, and have observed that the nadir usually happens between 4-5 hours afterward. Luckily, I work from home, so I watch his glucose closely at 9:30, 10:00 and 10:30 am. After that, the insulin starts working less and blood glucose starts to rise.
- There is more detail on curves later in this document with observations and lessons learned from monitoring my cat. At this stage in your journey, it is best to keep it simple.

Insulin application protocol (Test, Feed, Inject, Wait, Re-test periodically)

If you are working on the blood glucose curve, or not, there is an order in which to do things to get the correct results. In the simplest terms, you would test your cat's blood glucose, feed the cat, inject insulin, wait and retest.

- This order does not change, the reasons for each step may change, and the last step, re-testing may change in frequency and timing.
- Consistent order and timing between each step will ensure you get somewhat predictable results.

Test

It does not matter if you are using the ear-prick method or a CGM, you always need to know what the blood glucose level is before you proceed.

- You need to have a record of the time you did the test, as well as the resulting blood glucose level, either in a log book, or an app.
- In the beginning, you are recording this to determine the level of blood glucose you can safely administer the insulin dose at.
- Later on, you would use this level to determine if you can safely administer the insulin dose to the cat, or not.
- You always want to perform this test before feeding so that there is nothing influencing the reading.

Feed

Now that you have the results of the blood glucose level, you can feed the cat.

- You need to ensure the cat has eaten a sufficient amount to offset the amount of insulin you are going to give.
- You should have a minimum 15-minute wait for digestion of carbohydrates to start before giving the injection.

- In the beginning, you need to understand how much the cat needs to eat before giving the injection.
- As a general rule, it should be more than 50% of the food portion. Waiting for the cat to eat it all may not be possible. Try to aim for consistency.
- In the beginning, you need to record the time you fed the cat, what they ate and how much they ate to establish this threshold.
- Later on, you would hold off the injection until this threshold has been met, although recording how much of what was eaten in how long is very useful.

Inject

- Be sure to record the time the injection was given and the dose.
- If there were any issues giving the injection, be sure to record these as well. (I.e. "Cat moved, not sure if it completed properly.")

Wait

- In the beginning while you are determining the blood glucose curve, you will wait for a shorter time period, as specified by the vet. (I.e. 30 minutes, 1 hour, 2 hours, etc.)
- Later on, after the curve is established, you might wait for a longer time, basically aiming to re-test when the blood glucose should be at the nadir. It depends on how confident you are that the timing is right.
- If the nadir has been well established and you are very confident the cat will be at a safe blood glucose level, you may not re-test and just wait until the next injection. This is something you should confirm with the vet. I would always re-test at the nadir out of caution.

Re-Test

- If you hit this step, be sure to record the time and blood glucose level.
- If you are working on the blood glucose curve, then you would go back to the wait stage.
- If you see the blood glucose is low (less than 5 mmol/L) you also may want to wait for 15-30 minutes and re-test again so you can respond to possible hypoglycemia.
- If the curve is established and you are sure you are at the nadir, and the cat has a safe blood glucose level, you may want to wait until the next injection.
- This is a decision you must make from experience, with a recommendation from your vet.

Getting the Insulin dose right (Start Low, Go Slow)

This section will give you an awareness of how vets determine the appropriate insulin dose for your cat, it is intended to give you an idea of how it should be done so you can feel confident in their approach, not to replace their guidance.

If you do find your vet (like the first one I had) is not following this method, then I do recommend you find a new vet, like I did. I became convinced that this vet was using the wrong method and would ultimately end up killing my cat, so I found better care.

- The AAHA protocol is shown here as a "flowchart": <https://www.aaha.org/resources/2018-aaha-diabetes-management-guideline-for-dogs-and-cats/monitoring/>
- There are other sources for this protocol, they call it "Start Low, Go Slow", but to keep things simple, I will summarize it below.
 - "Start low" means that we start at a low dose of insulin and watch the nadir.
 - "Go slow" means we give at least 3-5 days to allow the body to adjust to the new insulin dose.
 - Certain insulins have more than 12-hour duration, so this gives the insulin time to "build up" in the tissues. Changing the dose too quickly can result in unpredictable results.
- Once, I had a dose working well for 9 days and then suddenly, my cat's glucose went very low (<4.0), so my vet will sometimes even go up to 11 days on a dose.
- I even had what I thought was a perfect dose that "overperformed" one day and dropped the blood glucose very low (<3.8) after 19 days, so you cannot be in a rush.
- If after the appropriate time has passed, and the nadir is over 8.33 mmol/L, the insulin dose is increased.
- If at any time the nadir is under 4.44 mmol/L, the insulin dose is decreased immediately.
- The idea is to get the best dose that avoids hypoglycemia while reducing clinical signs of hyperglycemia in your cat, improving quality of life for both of you.

Storage and Preparation of the insulin

Insulin is not something you can store on a shelf forever in any environment. It has a specific life and should be kept at a specific temperature in order for it to remain effective.

- I would recommend you read the data sheet that comes with the vial of insulin at least once. This will help you verify the instructions your vet gives you.
- When asked if I needed to agitate the ProZinc before I drew the dose into the syringe, I was told by my first vet I did not need to. The instructions on the data sheet clearly stated to slowly roll the vial to mix the insulin.

Storage

- The best storage temperature for all opened and unopened vials of all insulins is between 2°C to 8°C (36°F to 46°F).
- These insulins are biologically derived and highly sensitive to heat, light, and freezing.

Insulin Type	Duration After Opening	Notes
Vetsulin (porcine lente)	28 days	Must stay refrigerated; discard after 28 days even if unused
ProZinc (PZI)	60 days	Must stay refrigerated; discard after 60 days
Glargine (Lantus)	28–42 days	Can be stored at room temp ($\leq 25^{\circ}\text{C}$) for up to 28 days; refrigeration extends life to 42 days depending on brand
Detemir (Levemir)	42 days	Similar to Glargine; refrigeration preferred for full shelf life

You should discard and replace Insulin if:

Temperature Abuse

- **Frozen insulin:** Even brief freezing denatures proteins — discard immediately.
- **Overheated insulin:** Exposure to temps above 30°C (86°F), especially prolonged or repeated, compromises potency.

Visual Changes

- **Cloudiness or clumping in clear insulins:** (e.g., Glargine, Detemir) → discard. ProZinc should appear cloudy and will settle.
- **Color changes:** Yellowing, browning, or any discoloration is a red flag.
- **Floating particles or precipitate in any insulin type** → discard. ProZinc will settle and requires agitation before application.

Expired or Overused

- **Past unopened expiration date** → discard.
- **Exceeded in-use shelf life, or duration after opening** → discard.

Inconsistent Dosing Response

- If BG curves show unexpected resistance or erratic drops, and technique is confirmed accurate, consider insulin degradation.

Preparation

I can only speak as someone who uses ProZinc. I don't think this protocol will hurt with any other Insulin, but again, read the data sheet. It will give you the necessary instructions.

- A medication is subject to gravity, larger, heavier particles will collect at the bottom of the vial. This is what I see with ProZinc after a few hours in the fridge. I see the white particles settled at the bottom.
- Moderation of action is the key, mixing the active ingredients in a way that air (froth, etc.) is not introduced.
- The data sheet for ProZinc recommends rolling the vial, so turn the vial upside down, at a 45-degree angle and roll it gently, then I turn the vial right side up, incline it at 45 degrees and roll it again. Finally, I turn the vial horizontal and repeat. By that time, the insulin is pretty well mixed and is ready to draw the dose.

Application (the injection)

This was one of my biggest hurdles. I have had many injections and have some resistance to injecting myself, the thought of injecting another individual was a big hurdle for me to overcome. It really was one of my fears. What if I got it wrong?

- Do not inject insulin too close to the sensor if you are using one.
- The vet will more than likely show you their way to inject a cat, there are many ways to do this, it really comes down to what works for you. There aren't many ways to do it wrong.
- The concept is to form a "tent" of the skin, with a cavity above the muscle where you want the needle to enter. This is where you want to place the insulin dose. You do not want to inject the insulin in the muscle.
- One suggestion is to practice with an orange or lemon. There is a definite difference between the resistance you feel as the needle pierces the skin, then when it is past the skin and into the air cavity. The orange peel acts like the skin. This will help you know when you have it right.
- Some web sites recommend drawing back on the syringe and if you see blood, try another site before injecting the dose. My first vet pointed out that if I got used to the "transition", it would be better because drawing back can cause pain to the cat and unpredictable responses.
- If the vet offers to use a saline solution for you to practice with, do so. I did not and if I could go back in time, I would tell myself do it. It gives you a good first experience and would have gone a long way to increasing my confidence.
- The basic technique I was taught was to make a triangle with my thumb, forefinger and middle finger with my forefinger at the tip of the triangle. Grasp the skin, do not grab any muscle and lift up. This will form a tent of skin, with a divot under the forefinger. With the syringe at a 45-degree angle down, inject at the center of the divot.
- I also hold the syringe between my thumb and middle finger with the forefinger off the plunger until the needle has hit the cavity, then I use my forefinger to push the plunger down. If the cat moves, or resists, you cannot accidentally inject the insulin if your finger is not on the plunger.
- Your injection spot can be either on the hindquarters, the shoulder, or the scruff. You always want to change where you perform the injections, but it does not have to be a big change.
- I usually inject on the scruff. There is one reason for that: He sometimes will use his hind legs, or move the resist the injection. The reflex of relaxing when I make a tent on his scruff means he does not resist and makes it more likely the injection will be successful.
- Normally my forefinger is closer to his hindquarters, and this mean if the cat lunges forward, I may get an incomplete injection with some of the dose on the outside of his skin. To prevent that, do the reverse, have the forefinger facing the head of the cat so when it lunges forward the needle stays within the cavity.
- The choice of where and how to inject is yours to make. As long as you can make the tent.
- With all animals, empathy plays a big component, stay calm and focused and most of all, be confident. If you are nervous, hesitant, or stressed, your pet will respond, and never in a good way.

When not to inject

There are times you need to not inject the cat, the primary risk is hypoglycemia, which we have already discussed.

Cat does not eat

Your cat needs to eat to combat the onset of the effect of insulin. If it doesn't, you will likely watch their blood glucose plummet and risk hypoglycemia.

- The first vet did tell me not to inject if the cat did not eat, but eat how much and how fast? My cat nibbled on his food, I thought that was enough, but I was wrong.
- A normal healthy cat will almost always eat, maybe not all the food at once. My cat usually eats about 1/4 can, drinks some water, eats more in about 10 minutes and again later. You must learn to recognize this pattern and keep to it, and if it doesn't happen, ask what is wrong.
- An unhealthy cat may not eat, or as much if they don't feel well. I can't recommend what to do, but your vet has medications that can help. I always keep a couple doses in my fridge for these cases.
- My cat would sometimes rather not eat and starve if he doesn't get some flavor variety or doesn't like the taste of the food. A little bit of low carb, off the shelf pate mixed in the diabetic food will introduce that variety.
- Cats can also get hungry from environmental factors. He has a metal bowl, so I sometimes mix his food again, mound it in the center of the bowl and tap the spoon on the lip of the bowl twice to get the cat food off it. He usually gets a little hungry when that happens.
- It might also be worth a 15-to-30-minute wait (or an hour after medication) and see if the cat eats. It is always better to not give the injection than risk hypoglycemia.

Low glucose (below expected ranges)

I wanted to know about how this works and how to recognize it. The blood glucose curve helps you figure out this range and you don't want to gamble injecting the insulin if blood glucose is below it. It is better to let the blood glucose go high because you did not inject than have it go low because you took a risk.

- Let's use some simple numbers. Assume that is your expected observed glucose level before the injection is 20 mmol/L. Also, add the dose of insulin is 2.0u. Assume your nadir is near perfect at 6.5 mmol/L.
- If you see the pre-injection measurement at this time is 16.9 mmol/L and you give the same dose of insulin, the expected nadir will be below 6.5 mmol/L, hopefully closer to what I calculate at 5.5 mmol/L. It could be much lower, and you may have a condition where hypoglycemia is an issue.
- I have a CGM on him, and it stops registering blood glucose over 22.2 mmol/L, but recently his peaks have started getting into a measurable range. This has just started being a factor I can consider. What I do take into account is how long his blood glucose has been over maximum.
- The best solution is to wait 15 to 30 minutes and check again.
- I chose a different path, injected him with a lesser dose and in just a few minutes afterwards, before the food could take effect, his blood glucose hit the target and I regretted being impatient.

An alternative approach (requires math, how humans do it)

**** Warning - this might require some math. ****

My opinion is that not giving an injection leads to a greater risk of prolonged Hyperglycemia and resulting insulin resistance. This will impact the existing curve and it may take a lot of time and work to re-establish it. The administration of a smaller dose of insulin (1-1.5 units) can reduce the blood glucose and avoid the risk of hypoglycemia. The question is - how small or large a dose?

- The basis of my approach is in forgetting it's a cat and ask what humans do. A human will check their starting blood glucose level, calculate how much insulin they need to offset the calories from the meal they are about to eat, or have eaten, to get to the ideal blood glucose level and inject that amount.
- The same approach is what I use. I know certain things such as the optimal blood glucose level (23), the current blood glucose level (17), the dosage of insulin I give at that time (2.0u), and how low the nadir should be (6.5). The start of the formula looks like: $(\text{Actual glucose}/\text{Expected glucose}) \times \text{Insulin dose} = \text{Reduced dose}$.
- Another variable I need to add is how many kcals the amount of food the cat ate should give versus the ideal amount should be. I can't give a precise formula at this time; I am still figuring it out. So far though, my "best guess" keeps him from going hyperglycemic for long and hasn't shown any risk of hypoglycemia.
- If you think you want to do this, consult your vet first.

If the injection goes wrong

This is one factor you cannot work around. The problem is that an undetermined amount of insulin has been injected. Not knowing that, you cannot accurately know how much you need to make up for the loss.

- My cat has kicked the syringe and dislodged it; he has laid down, lunged forward and rolled over. All of these actions can pull the syringe from the pocket and waste the rest of the insulin. You cannot recover, but you can learn from the failure.
- Hopefully, part of the dose made it to the target.
- Do not try anything creative here, just accept it and move on.

When you are not sure you injected already

This happened to me once. I had a syringe ready and was going to inject him, and then the phone rang. After the call was done, I remembered filling the syringe, but did not remember injecting it and had no record to prove I did.

- This is why you record the injection immediately afterwards.
- Since there was doubt, I did not inject the cat until I found the syringe I already had ready.

Syringes - identification, gauge, etc.

U-100 and U-40 insulin syringes differ in insulin concentration, volume markings, and color coding.

- A unit on a U-40 syringe is 2.5x larger in volume than a unit on a U-100 syringe.
- Using the correct syringe for the insulin type is critical to avoid dosing errors.
- Both types of syringes can be purchased and delivered from Amazon for those who are having difficulty finding a good source.

- I prefer to use .3ml syringes with .5-unit markings and 31 gauge. The higher gauge seems to not irritate my cat as much during the injection.

Basic Properties of Insulin Syringes

- **Purpose:** Designed to deliver precise insulin doses subcutaneously.
- **Volume Sizes:** Commonly available in 0.3 mL (30 units), 0.5 mL (50 units), and 1.0 mL (100 units).
- **Needle Gauge:** Typically, 28G to 31G (higher gauge = thinner needle).
- **Needle Length:** Ranges from 4 mm to 12.7 mm depending on patient needs.
- **Graduation Marks:** Calibrated in insulin units, not mL, to match expected insulin concentration.

Feature	U100 Syringe	U40 Syringe
Insulin Concentration	100 units/mL	40 units/mL
Volume per Unit	0.01 mL per unit	0.025 mL per unit
Common Use	Human insulin	Veterinary insulin
Cap Color	Orange	Red
Labeling	Clearly marked "U-100"	Clearly marked "U-40"
Availability	Widely available in pharmacies	Often found in veterinary clinics

Using U-100 versus U-40 Syringes

Working to determine the proper insulin dosage, I found myself with a problem. I was using a U40 syringes and found that 2.5 units was too much and 2.0 units was too little. I asked my vet and was told we could always compound the insulin, basically diluting it. I wondered about the cost and investigated this solution and found that the insulin data sheet recommended against this approach.

- I found that the best solution was to use a U100 syringe for the U40 insulin (never do it the other way with a U40 syringes for U100 insulin.)
- The math is pretty easy. For instance, in my case, 2.2 units would be worth trying. So, 2.2 units * 2.5 (conversion factor) is 5.5 units in a U100 syringe.
- You can find a chart here: <https://www.raymed.co.uk/blogs/news/converting-u40-insulin-units-to-u100-insulin-syringes>

Hypoglycemia

We have been focusing on why high glucose levels are bad, but the opposite is true as well. Very low blood glucose is called hypoglycemia and at its worst will kill your cat.

- The vet (no longer my cat's vet) wasn't very helpful and only told me that if I saw him twitching and having seizures to rub corn syrup on his gums.
- It was one of my biggest fears that I inject my cat one day, go to bed and wake to my cat having seizures or worse.
- I have a CGM on him and the times he has been close to hypoglycemic, before the symptoms set in, the alarms warned me and I was ready.

What it is and why it is dangerous

Hypoglycemia in cats, low blood sugar, is a potentially life-threatening condition that occurs when blood glucose levels drop below normal, typically under 3.3 mmol/L (or 60 mg/dL)

- It is possible for a cat to have low blood sugar and not show symptoms, but the longer it continues, the bigger the risk and more likely you will see symptoms.
- Glucose is the primary energy source for the brain and body. When levels fall too low:
 - The brain is starved of fuel, leading to disorientation, seizures, or coma.
 - The body loses coordination and strength, which can rapidly progress to collapse.
 - If untreated, hypoglycemia can be fatal, especially in diabetic cats receiving insulin therapy

How to recognize it if you don't use a CGM

The only symptom I knew about was lethargy, but my cat sleeps a lot. How was I going to know the difference between hypoglycemia and normal behaviour?

Early Signs

- Hunger or restlessness
- Shivering or trembling
- Anxiety or confusion

Moderate Signs

- Weakness, lethargy
- Vomiting
- Stumbling or incoordination

Severe Signs

- Seizures or convulsions
- Coma
- Death (if untreated)

YouTube Videos

There are several examples on YouTube that you can use to train yourself to recognize this early. If you see these signs, get your kit ready, test the cat's blood glucose and be ready to respond.

<https://www.youtube.com/watch?v=2uOUmWiKRwI> (Unnamed)

- Most likely Early, a good candidate for Glucose and hi carb food with follow up monitoring. He should improve.
- 00:00 - He's behind the door, disoriented. His vision seems affect and cats will follow walls and surfaces trying to get to where they are going.
- 00:37 - His gait seems weird, almost drunk.
- 01:00 - He has passed the food bowl twice. She said she gave him honey, but he has no wet food? Is he looking for food?
- 01:30 - It is very quick, but look at the mouth, panting
- 01:45 - Panting is more visible

<https://www.youtube.com/watch?v=bN4ajPd-0PU> (ZiaoFiFi)

- Most likely Moderate, administer glucose and see if the cat will eat hi carb food. Monitor, and if no improvement, get to emergency.
- 00:00 - watch the eyes
- 00:09 - possible seizure, certainly muscle tremors
- 01:12 - pupils

<https://www.youtube.com/watch?v=7QfhqwQskLk> (Turtle Meeper, in a vet's care)

- Severe, if your cat looks like this, give glucose, get it to emergency immediately, call ahead and tell them what is happening.
- 2:19 - watch the ears
- 2:24 - watch the mouth, panting
- 2:49 - whiskers and mouth, back leg seizures
- 3:02 - pupils

If you use a CGM

All CGM's have a phone app you can use for monitoring. These apps have alarms to tell you when blood glucose goes under a certain level.

- Usually, the low threshold is 3.8 mmol/L, giving barely enough time to respond with glucose followed by hi carb wet food.
- Alarm thresholds can be customized.
- I turn off the high glucose alarm.

Treatment and response

The response can vary based on the amount of warning you get and how many symptoms the cat is showing and how severe they are.

- Because I work from home and am usually watching the CGM app, I can be much more proactive.
- If you are using the ear-prick method, then ensuring you check blood glucose at close to the average nadir time can alert you to danger.
- Cats can generally go under 3 mmol/L blood glucose, but not for long. There may be a response from the liver (a release of glucose) that may save the day, but you cannot count on it, the response may already have happened and had little effect.
- If you see the symptoms above, assume the worst, start with the glucose and follow up with high carb food. Monitor every 5-15 minutes and if there is no improvement, get to emergency and call ahead to tell them you are coming and why.

Corn Syrup (Glucose)

Glucose is an effective and fast acting solution, but it is only short lived. It can buy some time for a longer lasting response to start acting. Crown (brand) white corn syrup is available in the baking section of most grocery stores. It is mostly glucose and very effective.

- It will give an almost instant glucose spike lasting only a few minutes, the effects wear off quickly which is why I usually follow up with hi carb wet food.
- You may have to apply glucose multiple times, one day when I injected insulin when I should not have, I had to give him glucose 7 times in a 5-hour span. He lived, and I learned.
- My cat does not like glucose. If the alarm in the GCM goes off, he knows what is coming and he will fight as hard as he can to avoid the inevitable, ruling out the gum rub. Usually, I get coated with the stuff, and more gets on me and his fur than actually gets on the gums.
- There are two ways to administer it, by pouring some on your finger and rubbing it onto the gums, or by using a syringe. The blood vessels under the lips and tongue absorb glucose more quickly than the stomach.

Gum rub

- You need to use a lot, so if you need a lot, apply it more often in smaller amounts.
- The application is simple if the cat does not fight. Put a good dose on your finger tip, hold the cat and rub it on the gums on the outside of the mouth, just under the lips.
- Do not stick your finger in the mouth. If the cat fights it, or has a seizure, you might get severely bitten.

Syringe

Pharmacies usually have large syringes that are used to give liquid medicines orally to babies. I have one of those.

- To make the corn syrup easier to use in the syringe, I dilute it in a little warm water in a glass and then load the syringe from that.
- Holding the cat, I place the syringe opening between the lips and the gums at the back of the cat's mouth near the molars and squirt the mixture in the gap between the lips and gums.

Diabetic wet food

Diabetic wet food usually is the gentlest of responses. It is usually formulated for the carbs to act quickly, but may have limited effect.

- I start with a feeding of diabetic food if blood glucose is under 5 but above 3.8 mmol/L. I would rather be less than perfect than have to take my little buddy into emergency.
- You should see some response within 5-10 minutes. It won't be as strong a response as high carb wet food, but may be sufficient to do the job without causing too huge a spike in blood glucose.
- I don't know how my cat knows it, but he will usually eat just enough to stop the downward trend.

High carb food

- This is usually my second level of response, and would be if I had to administer glucose as well.
- It can take 5-10 minutes to take effect and will certainly cause a large spike in glucose.
- The downside is that it is tasty and then I have to fight later to get him to eat the diabetic wet food.
- The spike may affect his blood glucose curve for several days following.

Have a kit handy

There are times when the cat is more likely to go hypoglycemic, but in order to properly respond, you need to have the requirements nearby.

- Here is what I have in my kit:
 - 2 cans of Fancy Feast (any poultry will do) Shredded in Gravy (between 12-14% carbohydrates)
 - A small bottle of Crown Lily White Corn Syrup
 - A baby medicine syringe
- Hopefully, you will never need it, but if you don't have it, then expect a large emergency vet bill. An overnight stay with a glucose drip and constant monitoring is way more expensive than being ready to respond.

Glucose toxicity and Insulin Resistance

These two terms you will hear a lot because they are basically at the heart of diabetes. They create a vicious cycle that you need to manage with insulin injections, and hopefully once you get this cycle under control, your cat's health will improve.

Glucose Toxicity

A state where persistently elevated blood glucose levels damage pancreatic β -cells and impair insulin secretion.

- Glucose toxicity in cats is primarily driven by sustained hyperglycemia, not transient spikes.
- Chronic elevation impairs pancreatic beta-cell function, reducing endogenous insulin production.
- Reversal is possible with tight glycemic control, often within days to weeks if insulin therapy is effective.
- Stress-induced hyperglycemia (e.g., vet visits) can temporarily spike glucose > 16.7 mmol/L, but doesn't cause toxicity unless sustained.

Blood Glucose Level (mmol/L)	Duration of Elevation	Risk Level / Notes
> 16.7 mmol/L	Several days	Onset of glucose toxicity; beta-cell suppression begins
> 22.2 mmol/L	Persistent	Significant insulin resistance; poor response to exogenous insulin
> 27.8 mmol/L	Chronic	Severe glucose toxicity; risk of ketoacidosis and long-term damage

Insulin Resistance

- A metabolic condition where muscle, fat, and liver cells become less responsive to insulin.

How are they related?

- Insulin resistance causes elevated blood glucose.
- Elevated glucose leads to glucose toxicity, damaging β -cells.
- Damaged β -cells produce less insulin, worsening insulin resistance.

Diabetes Risk factors

Breed: Burmese cats in the UK, Australia, and New Zealand have a significantly higher incidence—up to 3.7× that of other breeds.

Age: Risk increases significantly in cats over 7 years old

Body Weight: Overweight cats are up to four times more likely to develop diabetes.

Gender and Spayed/Neutered: Male cats, especially neutered males, are at higher risk.

Diet: High-carbohydrate dry food is a major contributor.

Steroids and Medication: Drugs like prednisone can induce insulin resistance or trigger diabetes.

Chronic Disease

- Pancreatitis
- Hyperthyroidism
- Acromegaly
- Cushing's disease (hyperadrenocorticism)
- Chronic infections

Remission

Diabetic remission in cats means that a previously insulin-dependent cat maintains normal blood glucose levels without insulin injections for an extended period, typically at least four weeks. It's a clinical recovery, not a cure, and requires ongoing monitoring.

What are the stats for remission by insulin type?

Insulin Type	Remission Rate	Notes
Glargine (Lantus)	Up to 84–100%	Highest rates when started within 6 weeks of diagnosis
Detemir (Levemir)	Similar to glargine	Long-acting, stable profile; excellent for tight glycemic control
PZI (Protamine Zinc)	~25–40%	Intermediate-acting; less consistent control than glargine
Vetsulin (Caninsulin)	~15–30%	Veterinary insulin; lower remission rates, especially with delayed start

- Source document here: https://felinediabetes.com/FDMB/attachments/management-of-diabetic-cats-roomp_rand-peer-reviewed-study-pdf.31473/

How does nadir affect remission?

The nadir is a critical marker of insulin effectiveness and remission potential.

- Tight glucose control (including a proper nadir) helps reverse glucose toxicity, allowing pancreatic β -cells to recover.
- Optimal Nadir Range for Remission
 - Home monitoring: 65–117 mg/dL (3.6–6.5 mmol/L)
 - In-hospital monitoring: 72–160 mg/dL (4.0–8.9 mmol/L)
 - These ranges are tight enough to reverse glucose toxicity without causing hypoglycemia.
- Source:
https://assets.ctfassets.net/4dmg3l1sxd6g/2lowxj0w44BtRYxV9SXXjO/181de4070f33558c78850868ddc63888/cap_in_ducing-diabetic-remission-in-cats-redefining-hemorrhagic-gastroenteritis_0-18641-article.pdf

What are the other factors that contribute?

Factor	Impact on Remission
Early insulin initiation	Strong positive effect
Low-carb canned diet	Strong positive effect
Weight loss in obese cats	Moderate to strong positive effect
Pancreatitis or acromegaly	Negative effect
Corticosteroid use	Negative effect
Burmese breed (some regions)	Variable; may affect risk and response
Owner monitoring & compliance	Strong positive effect

What signs will you see?

If you are not properly monitoring your cat's blood glucose, the first sign of the onset of remission would be the cat going hypoglycemic.

If you are not using a CGM and relying on the ear-prick method, you will catch the low pre-injection glucose levels, but may miss the nadir.

Pre-injection low glucose levels

If the cat has a lower-than-expected blood glucose level before the injection time, this may be a sign that the pancreas is recovering.

- This is why I try not to miss injecting the cat, a smaller insulin dose is better than the risk of undoing all the progress by allowing the cat to go hyperglycemic for even a short time.

Lower nadir

If you notice the nadir going lower than normal, then something has obviously changed and one option is that the pancreas is starting to recover.

The dilemma

There are just so many variables in cat diabetes that could be causing either of the two blood glucose measurements to wander and change.

- If the signs persist, and it is not a one-time variation, then your cat's health probably is improving.

What do you do if you see the signs?

This first thing you should do is contact the vet, bring the blood glucose log and get their advice.

- One thing they can do is the Fructosamine test which can verify if the blood glucose control is improving.
- You will probably need to re-establish the blood glucose curve and adjust the insulin dose.

Monitoring

Blood Glucose level monitoring is extremely important with a diabetic cat. There are advantages and disadvantages to each method and what works for me may not be the method you choose.

- I recommend watching this (it really helped me): <https://www.youtube.com/watch?v=r6B9NUwsmaA> (Purdue Veterinary Medicine: How to Monitor Insulin Treatment for your Diabetic Dog or Cat)

Urine glucose

If you suspect you have a diabetic cat, you can confirm this fairly easily by testing for glucose in the cat's urine at the litter box. If you have multiple cats, you will need to watch the litter box and make sure you have urine from the correct cat, but it isn't difficult.

Cat litter (glucose)

I have personally never tried the cat litter, but would be tempted to use it periodically if my cat ever went into remission.

- There are several brands of cat litter that will change color with the presence of glucose in the urine.
- In Canada, a popular product with many pet product retailers is "OdourLock maxCare":
<https://odourlock.com/en/products/maxcare/>

Test strips (glucose and ketones)

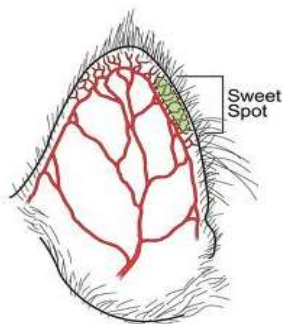
You can find many urine test strips for cats for conditions such as glucose, ketones, UTI's, etc. at many pet product retailers, either online or in store.

- These would work if you have one cat. You can use a hydrophobic (doesn't absorb liquid) cat litter, the urine will collect in the bottom of the litter box.
- I have read that you can take a sample of clumping cat litter, mix it with water and use the liquid as a urine sample.

Blood Glucose Monitor

These devices generally require a small blood sample, usually taken from the ear. They are very accurate when properly calibrated.

- There are various brands, most vets recommend the Zoetis AlphaTrak 3. One of its strengths is that requires a smaller blood sample than the other brands. (<https://www.zoetispetcare.com/products/alphatruk>)
- All of these devices require test strips and need to be calibrated with each new batch of test strips.
- Using a lancet, you prick a specific area on the ear where there are many capillaries, you don't poke the main vein. These heal quickly, but you need to rotate to a new site for the next sample.
 - Lancets come in many gauges, I prefer the thinner 31 gauge since despite what I have read, he can feel it, and feels it more with the 28-gauge lancet. He will usually flick his ear when that happens, losing the sample and forcing me to get another.
 - I don't use the Lancing Device, but rather use a cotton makeup remover pad folded in half on the inside of the ear in one hand, and use the lancet on the outside of the ear with the other hand.
 - A little "massaging" near the site with the cotton pad still against the inside of the ear will produce a large enough sample.
 - With the device on, I insert the sample strip, touch the sample with the strip and record the reading. This can be tricky due to a timeout, you need to be fast.
- There is an app you can install on your phone, with the device paired via Bluetooth to maintain a log of readings.
- Strips, calibrating solution, lancets and everything you need are available for delivery by Amazon if you have mobility issues.
- While accurate, you can only test so often and there is always a risk that you can miss the nadir, or an unexpected event in the blood glucose curve.



CGM Sensors

CGM means “Continuous Glucose Monitor.” I find that manual blood glucose monitors and CGMs typically have different uses with pets, but this is not the same with humans. A CGM in humans can link to an insulin pump with automatic dosing in real time to keep blood glucose levels within range, but a cat relies on a human to do the work of insulin dosing twice a day.

Core Components of a CGM System

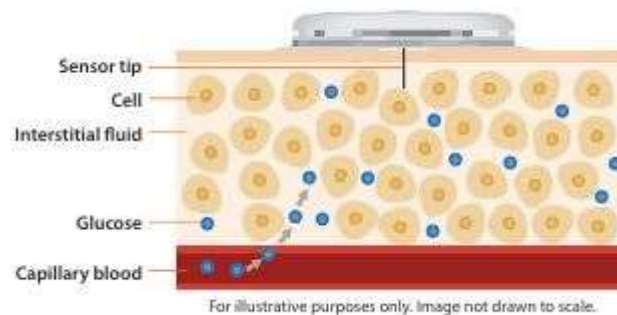
- **Sensor:** A tiny filament inserted just under the skin (usually on the abdomen or shoulder) that detects glucose levels in the interstitial fluid, not directly in the blood.
- **Transmitter:** Attached to the sensor, it wirelessly sends glucose data to a receiver, smartphone, or smartwatch.
- **Receiver/Display Device:** Shows glucose readings in real time, often with trend graphs and alerts for high or low levels.

Why it's better than traditional Blood Glucose Monitors.

- **Real-Time Monitoring:** Unlike traditional glucometers that give a single snapshot, CGMs provide a continuous stream of data.
- **Trend Analysis:** You can see how your glucose responds to meals, exercise, stress, or sleep.
- **Remote Monitoring:** Some systems allow caregivers or healthcare providers to view data remotely, which is especially helpful.

Interstitial versus blood measurements

Understanding the distinction between interstitial fluid glucose and blood glucose is key to interpreting CGM data accurately. Interstitial measurements are taken from microscopic gaps between cells or tissues where fluid exchange occurs.



- Interstitial fluid glucose lags behind blood glucose by approximately 5 to 15 minutes. This delay becomes significant during rapid glucose changes (e.g., after eating or insulin injection), potentially leading to misleading readings.
- Because of the lag, CGMs may not alert users to low blood sugar quickly enough, especially during sleep or after insulin administration.
- Users may act on outdated data, risking hypoglycemia or hyperglycemia if insulin is dosed based on interstitial readings during fast-changing conditions.
- CGMs may be less accurate during periods of rapid glucose rise or fall, such as during intense exercise or stress.
- Accuracy can be affected by where the sensor is placed—fatty tissue, hydration levels, or inflammation can alter interstitial fluid composition.
- Lying on the sensor (e.g., during sleep) can cause falsely low readings due to restricted interstitial fluid flow.

Accuracy versus frequency

Blood glucose monitors are much more accurate than CGMs. I have seen differences between the two of 1-2 mmol/L between readings taken at the same time by both devices.

- CGMs excel at being able to provide data every minute. Using a blood glucose monitor at this rate is impossible. The poor cat's ear would not deal with it well. It does better at showing trends.
- It is a lot harder to miss a nadir with a CGM, with a blood glucose monitor it's pretty easy to miss it.
- I find that using both devices together works extremely well, especially if the sensor is recalibrating itself, or otherwise is unable to produce a reading.

Convenience versus cost

The choice I made for the primary glucose monitoring was based on convenience as well as cost.

CGM

I can buy a Libre 2 sensor at my local pharmacy, attach it myself without the need of the vet at a cost of about \$110 (Canadian) per device. This includes the cost of:

- 50u Syringe (used for Vetbond application)
- Skin Tak wipe
- 3M Vetbond
- I get constant readings and don't need a vet appointment.
- I don't have to fight with the cat at 5am when he'd rather be fed than having his ear pricked.
- Each sensor, if properly applied, and if the cat does not remove it, usually lasts about 10-11 days.
- Using LibreView, I will always see the nadir.
- Using the app, and if I am in range, I will get an alarm before a Hypoglycemic event occurs.
- In an ideal month, this will cost me about \$330 (Canadian)

AlphaTrak 3

After the initial cost of the AlphaTrak 3 kit (\$150 Canadian), I can do a single test for about \$2.85, but not every one will be successful, so testing before feeding, re-checking the nadir, say 4 tests per day, it works out to \$11.40 a day. This includes:

- Alcohol wipes
- Cotton makeup remover pads
- 31 ga Lancets
- Test Strips
- Control Solution.
- At an average 30 days per month, this works out to \$342 (Canadian).
- If something goes wrong, and depending on the cat and your experience this chance will reduce, you will need to use a new test strip, about \$2.50 (Canadian) each.
- Since you only get 4 readings a day, you may not catch the nadir or a hypoglycemic episode. In either of these cases, you will likely use another 3 tests.
- You could easily miss a hypoglycemic event if you are not watching the cat, say when you are asleep.

Dexcom

One CGM is the Dexcom G7 ([Dexcom G7 Continuous Glucose Monitoring | Dexcom](#))

- I have not heard of any vet recommending the device, but have seen posts on the Feline Diabetes Forums that some cat owners are using it.
- I have not tried it, so cannot comment on this product.

Freestyle Libre

If you ask for a CGM, the vet will recommend the Freestyle Libre. ([What Is FreeStyle Libre | Products | Abbott](#))

- Abbott offers a free trial unit to get people with diabetes started. A prescription or diagnosis does not seem to be a requirement. Unfortunately, I used my cat's name and age when I created the account and had to provide parental consent. I doubt that qualifies for a free trial unit.
- Libre sensors do have warranty replacements for sensors that come dislodged prematurely, but this does not apply to sensors used on cats.
- The Freestyle Libre was used in a scientific cat diabetes study, hence why it is recommended by vets.

Sensor Errors

The biggest complaint about the Libre is the unwelcome "Sensor Error" that happens. Sensor errors normally occur when:

- Blood glucose levels fluctuate rapidly
- Glucose levels are abnormally high or low
- The sensor is poorly attached or coming dislodged.
- Inflammation at the attachment site is hindering accurate measurements.
- This is because the sensor is running diagnostics and other actions. Remember that these sensors are made for use by people and generally, human blood glucose does not fluctuate much when insulin is in use.
- It usually clears up after about 10 minutes unless the sensor is loose or dislodged.
- This can be very inconvenient since people get used to always getting readings. Not knowing causes stress. When blood glucose levels are dangerously low, it can be extremely stressful.
- I find that "spikes" caused by releases of glucose by the liver seem to be the main cause.
- If the error occurs when glucose levels are in a safe range or rising, they can be tolerated. You have time for the sensor to recover.
- If they occur when glucose levels are very low, I take the safe route and give the cat a bowl of diabetic wet food. Even if it is not needed, it is better to act than not. My cat usually eats just as much as he needs to counter any danger.
- If sensor errors are often and persistent, I replace the sensor.

Signal Loss

The 6-meter range of Bluetooth means the sensor must be near the phone. If the cat wanders out of range, then the app reports "Signal Loss".

- You can "scan" the sensor by holding the phone within an inch of the sensor.
- This can be used to retrieve up to 8 hours of data.
- This is done using Near Field Communications (NFC), the same mechanism that provides tap to pay services.
- If the signal is regained, the app will resume displaying readings, it is not necessary to always scan the sensor.

Hi Readings

The Libre sensors cannot provide measurements when the blood glucose is over 22.2 mmol/L. All the app shows for a value is "HI".

- This can be a problem for severely diabetic cats. As you regain control over blood glucose levels, this ceases to be a problem as the peak glucose levels lower.

Alarms

The Libre 2 app has three configurable alarms. Humans will likely use them all, but being able to turn them off and set the levels the alarms are triggered is very useful.

- **Low Glucose:** Usually set at 3.8 mmol/L, I prefer to set this to 4.5 so if the cat's blood glucose levels go low while I am asleep, I have extra time to wake up and get ready to deal with a possible hypoglycemic event.
- **High Glucose:** Usually set to 13.3 mmol/L, I turn this alarm off. It doesn't have much use for a cat.
- **Signal Loss:** This alarm triggers when the app is out of range of the sensor. I leave this on. All other alarms will not be triggered in the case of signal loss, so this is useful.

Calibration

The Libre 2 sensor comes pre-calibrated. I did some research on the internet and found an article that described how Abbott calibrates these devices.

- A single sensor is selected from each manufacturing batch.
- This sensor is then calibrated.
- The resulting calibration values are applied to all the sensors in this batch.
- It's probably one of the reasons why the readings from these devices are not perfectly precise.

LibreLink (Libre 2, Libre 3)

The Libre 2 sensor isn't useful without the Libre 2 app.

- There are separate versions of the app for Apple and Android phones.
- You will find it in your device's app store.
- The Libre 2 and Libre 3 sensors have different apps.
- The Libre 3 app may not be available if the Libre 3 cannot be purchased in your country.
- The manual, videos and tutorials for the app are here: [Tutorials & Downloads](#) | [Freestyle Libre](#) | [Abbott](#)

Supported devices

Before considering the Libre 2 sensor as a solution, you should check your phone to make sure it is compatible.

- Not all phones will support this app.
- Be sure to also check the operating system version compatibility.
- You can find the latest information here: <https://www.freestyle.abbott/content/adc/freestyle/countries/ie-en/support/compatibility-guide.html>
- I couldn't understand why I could not find the app in the Google Play Store until I realized I needed to upgrade my phone. That almost equalled the bill from the emergency vet.

Required cell phone capabilities

There are two critical features that the app uses to communicate with the sensor.

Near Field Communication (NFC)

NFC is the network capability used to provide tap to pay functionality.

- This is needed to activate the sensor and pair the phone to the sensor.
- It is also used to scan the sensor, which can retrieve up to the last 8 hours of readings if the signal is lost.

Bluetooth

Bluetooth provides a network communication from the sensor to the phone. This provides:

- Real time updates for blood glucose levels.
- Alarms are enabled.
- Bluetooth range is 6 meters.

Quick Notes

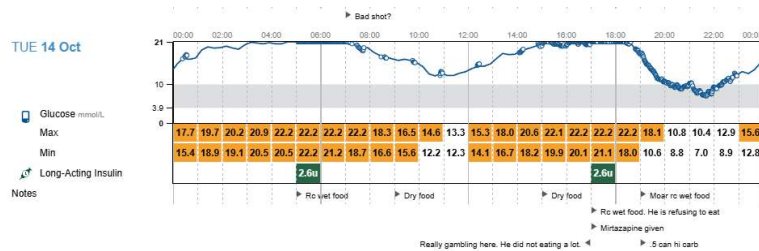


- A. This gives you the menu to switch between graphs and views, access settings, etc.
- B. This is the current blood glucose reading.
- C. This is a trend indicator. Do not take this too seriously. An arrow pointing straight down does not always mean the blood glucose level is going down fast. You need to watch the blood glucose reading over time to be sure.
- D. These notes are invaluable. You can record feeding and insulin dose and administration times which are useful in other tools. If you record what you do or observe immediately, then you have no doubt you did it. Tap on the icon to view or edit. These notes can help the vet understand what is going on if they view the data remotely. This is your log.
- E. Here are the glucose level readings visualized in a curve. This is what you need to focus on, is the trend going down fast, holding steady, does it match how fast the readings are changing? With the display scaling, it is hard to figure out in the beginning, but you get used to it over time.
- F. Start and end times of the observations.
- G. Click this to add notes. Cat insulin should be recorded as Long-Acting Insulin in units, record meals, grazing and other observations as comments.

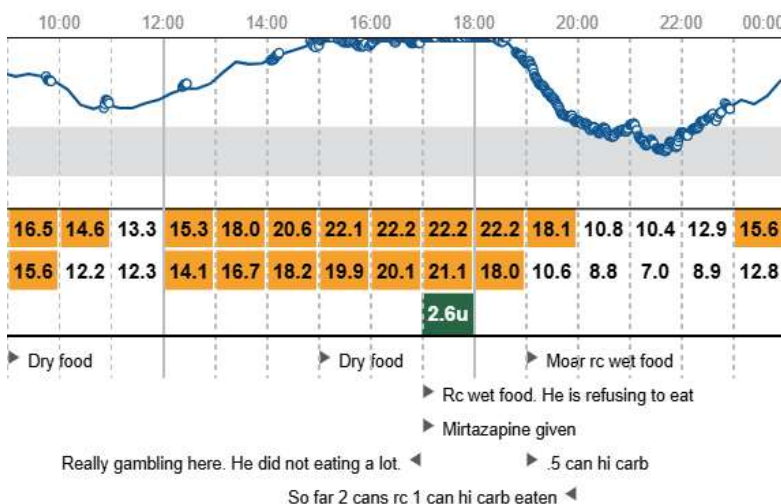
Importance of adding notes

Notes are a great time saver; this is your log. If you record them properly, they can be viewed by the vet and you do not have to remember the past. Let's take this day as an example.

- This snapshot is from LibreView which is commonly used by vets to access your cat's data.



This is the view zoomed in (with a timeline from memory):



- It started when he refused to eat and I gave him Mirtazapine.
- It does take an hour to take effect in tablet form, and I am glad it did.
- Unfortunately, I injected him hoping it would work faster and both drugs took effect.
- I can't really say what effect it had except he got really hungry, a good thing.
- He ate another can of diabetic wet food
- Then he ate ½ can of high carb food
- He finally ate let other ½ can.
- He still did not go hypoglycemic.

Here is the same day in my Power BI report:



- There are four panes, Insulin, Notes, Details and Time in State.

Insulin – How many units were given.

Notes – Observations and feeding details.

Details – Average glucose readings at that day in 10-minute intervals.

Time in State – How much time (percentage) the glucose levels have been in which range.

- All entries are sorted by the date and time they were entered.
- You can follow the timeline of the events from the time they were added.

LibreView

If you want to look at historical data, with different reports from your home PC, LibreView is the tool to use. Your vet will also request that you allow them access to this data. No vet should ask for snapshots from your phone, snapshots of LibreLink simply do not have enough information for diagnostic purposes.

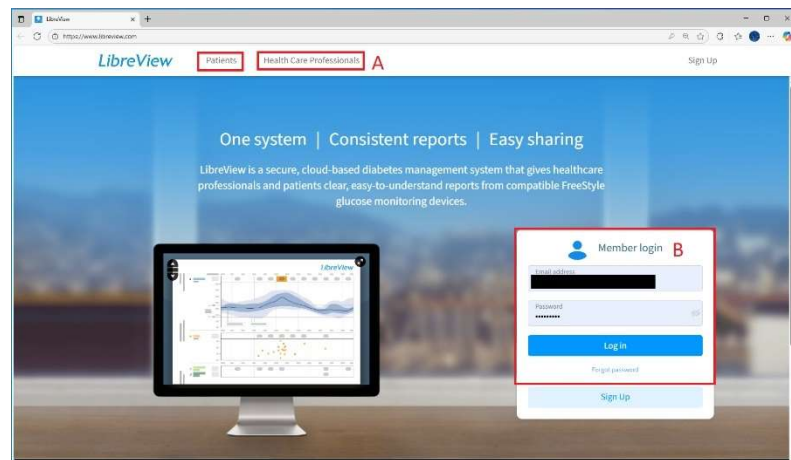
- LibreView does not cost any money.
- This is the system you created the account for in the Libre 2 app on your phone.
- You can access all your historical data from this system.

URL

The URL is: [LibreView](https://www.libreview.com/) (https://www.libreview.com/)

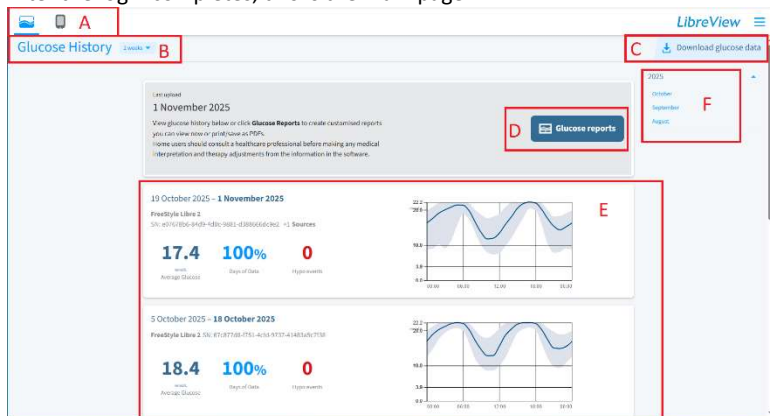
Walkthrough

The first page is the login page.



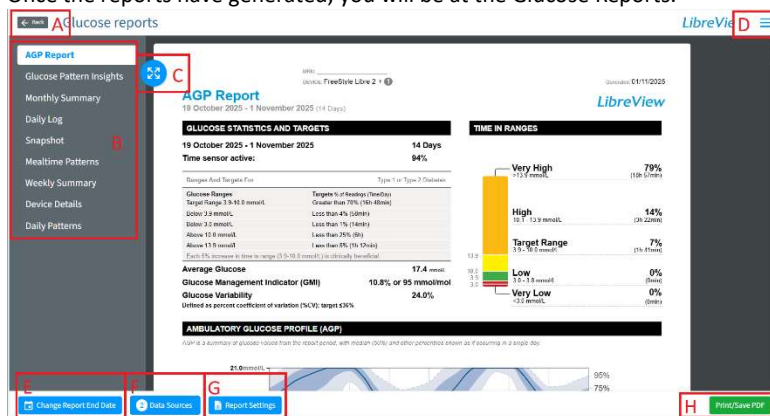
- Here you select the login type. By default, you are already at the "Patient" login page.
- Enter the email address and password you used to create the phone app account.

After the login completes, this is the main page.



- A. Unless you have a Libre 2 monitor, you don't need to upload data.
- B. Here you can select how many days of data each pane of the glucose history will show.
- C. Click here to download a .csv file containing all your glucose data with notes.
- D. Click here to proceed to the reports
- E. Glucose history (summaries)
- F. Quickly navigate to the glucose history by month.
 - You want to click the “Glucose Reports” button.

Once the reports have generated, you will be at the Glucose Reports.

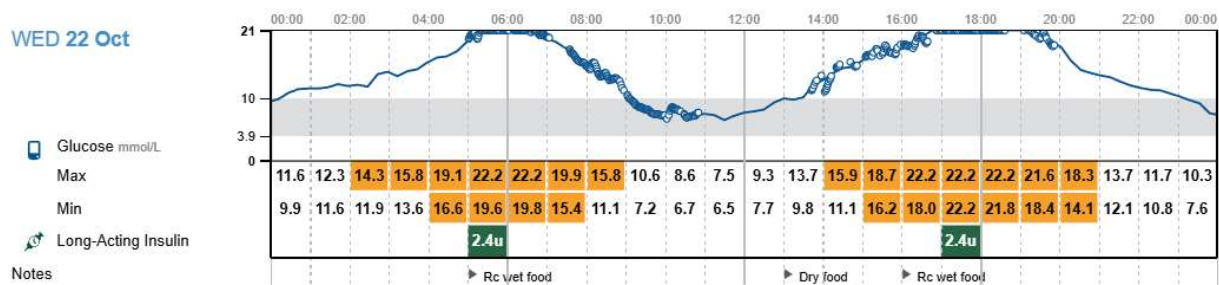


- A. Click here to go back to the glucose history.
- B. Use this menu to jump quickly to any report.
- C. Click here to hide the left pane.
- D. Click here to go to the settings menu
- E. Click here to change the end date for the report. This can allow you to go back and view historical data.
- F. Click here to see the sensors the report data came from.
- G. Click here to customize report settings.
- H. Click here to print/save the report to a PDF.

- The report thresholds are set for human patients. You cannot change them to accurately report glucose for a cat; the application will not allow that.
- Most of the reports are not really useable as a result, we are going to skip to the useful parts.

Daily Log

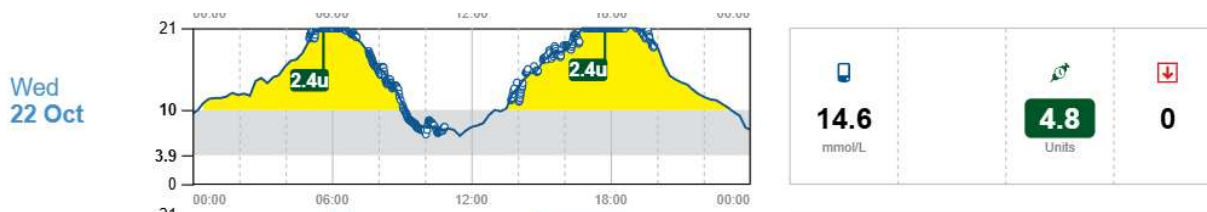
The daily log is the report you will use the most. The legend on the left explains what data is displayed.



- The data is summarized hourly. There is no mechanism to query data with better granularity.

Weekly Summary

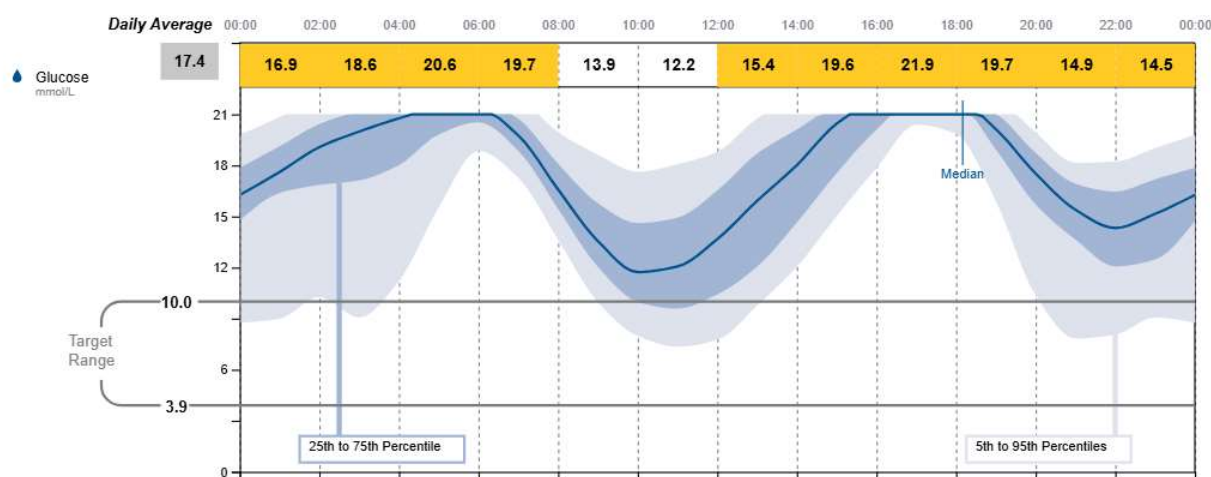
The weekly summary is less useful.



- The one useful piece of information is the average glucose (14.6 mmol/L). This is a good indicator for glucose control. The lower the number, the better.

Daily Patterns

This report is basically a scatter graph of glucose levels over the past two weeks providing stats on average glucose at specific times of the day.



- This chart is from a poorly controlled cat.
- In theory, we want the darker blue line (25th to 75th Percentile) within the target range as much as we can.

LibraLinkUp

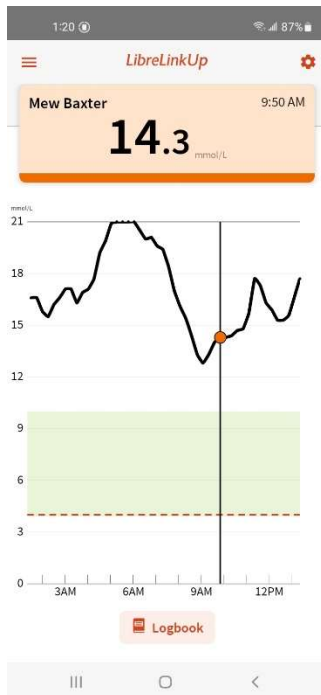
You can only have one phone paired to the sensor. LibraLinkUp allows you to share the data with other friends and family members on their phones. It is also very useful for extending the range of monitoring with one phone near the cat to receive data and another phone displaying near real time data and alarms.

- Say you don't work at home, but you want to monitor your cat while at work. Here is how to do it:
 - The phone with the Libre 2 app should not be your main phone.
 - Find a compatible phone you can use, a location where it can be within Bluetooth range of the cat and install the Libre 2 app on it.
 - Get a second email address if you don't have one you can use.
 - Invite this email, install LibreLinkUp on your main phone using this second email address after the invitation is received.
 - Now you can monitor your cat while at work so long as the cat remains in Bluetooth range of the first phone.

To share with friends and other devices

Instructions for sharing and generating the invitation is here: [Getting Started With LibreLinkUp](https://www.librelinkup.com/articles/getting-started) (https://www.librelinkup.com/articles/getting-started).

Limitations and strengths



- One nice feature is the ability to “swipe” the glucose curve graph to see glucose levels and times.
- To continue monitoring at the current time, you can tap inside the orange area to the right of the glucose reading.
- Insulin and notes are not displayed.
- Alarms can be customized by tapping the gear icon on the top right of the screen.
 - Turn off the “High glucose” and “No recent data” alarms.
 - Turn off “Glucose Reading” to stop notifications for when the sensor is scanned.
- The phone requires an Internet connection so it can connect to the data source.

Where to buy the sensors

You can buy Freestyle Libre sensors at any pharmacy. This is where you will find them in stock, and at the lowest price.

- I have yet to find a vet that has these in stock, they can sometimes take a couple of days to order them.
- It is recommended that you always have two on hand. If the application of a new sensor goes wrong, then you can use the second one.
- You will not always be able to plan when you need to replace one. I have had to replace one at 3 am, not a pleasant experience for me or the cat.

Libre 3 versus Libre 2?

The Libre 3 has just become available in Canada, so now there is a choice of sensors you can use.

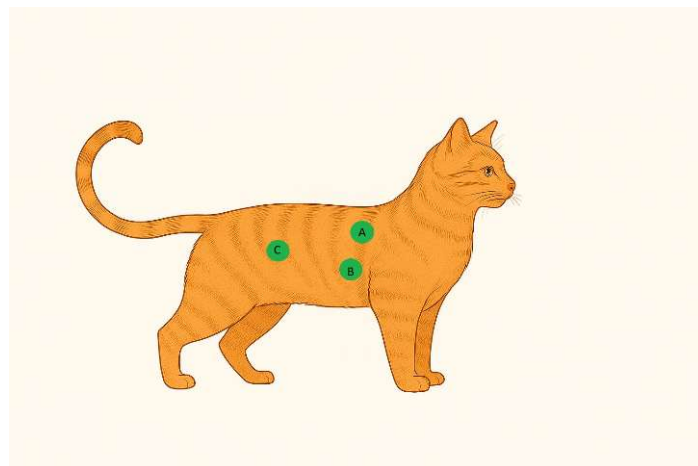
- A Libre 3 sensor is smaller than a Libre 2.
- The Libre 3 and Libre 2 have the same specifications.
- The Libre 3 costs about 25% more than the Libre 2.

Application of the sensor

If you do not think you can apply the sensor, you can always have the vet do it, but you may not get an appointment quickly. It is actually pretty easy to do and becomes much faster and easier as you get experience.

Possible sensor locations

There are three common locations for the sensor. You need to select a place where you will not hit a bone, it has some cushion and move as little as possible. The less a sensor moves, the lower the chance it will get dislodged.



A. This is the best location I have used. It is slightly above and behind the shoulder. It doesn't move much and is hard to lick. It can be easily covered by a bandage and/or a halter. The downside is that the cat can use their hind leg to kick the sensor off.

B. Just behind the top of the leg, it can move and can get exposed at the top of the leg in a halter. It is pretty hard to lick. Too far back and you may hit a rib, wasting a sensor.

C. In the middle of the thorax, just behind the ribcage. The sensor can end up anywhere from the ribcage or the top of the hind leg. It's hard to kick off, but very easy to lick.

- I have heard of people placing a sensor at the base of the neck between the shoulder blades. This is too close to my preferred injection site and cannot be duplicated on the other side of the cat. It's good for one use.
- Because I rely on a CGM, and because it needs to be changed about every 10 days, I need another ideal spot on the other side of the cat to allow time for the old site to heal.
- You cannot put a replacement CGM in the same location as the last one.

Prepare the site

In order for the sensor to stick, you must remove as much hair as possible, as well as dander. I can tell you, trying to get something like a Libre cover is almost impossible on my cat, it's almost like his skin is made of Teflon.

- In the beginning, you can use hair clippers, they work well, but do take off more fur than you need.
- Once I have a good spot selected, I use a beard and mustache trimmer. It does pretty much the same job, but for a smaller area.

Sterilize the site

The sensor has a small filament that will be implanted under the skin. To remove the dander and sterilize the application site to avoid infections, use the two alcohol wipes included with the sensor and allow the site to dry.

Assisting adhesion

Having the sensor properly adhered to the cat's skin will help prolong the life of the device. It took me many attempts to get to the point where I could get a sensor to last a reasonable amount of time without falling off, especially if it moved a lot.

3M Vetbond (Tissue glue)

The most common method you read about is to use "super glue" by placing one drop in each cardinal direction – north, south, east and west. **DO NOT use regular super glue, it can irritate the cat's skin.** You need to use a tissue adhesive such as [3M™ Vetbond™ Tissue Adhesive, 1469SB, 0.1 oz \(3 mL\) | 3M United States](#).



- This is a Libre 2 with the adhesive fabric removed. I have added blue dots for the cardinal locations. You need a fair-sized drop. Don't get the glue on the filament.
- I add another drop of adhesive between these locations (total of 8).
- After about 10 days, the adhesive will have been absorbed by the skin.
- Note the ridges in the plastic, these will show up as white lines in the adhesive fabric. Put the adhesive on the outside edge, where there is a ridge.
- You will need to peel the fabric from the applicator using forceps or a fingernail when you remove it.
- I use a 40u syringe, opening the top of the vial of adhesive, filling about 1/8 of the syringe and then using it to precisely place the adhesive on the Libre 2. It's more precise and you don't need a lot of the adhesive.
- You can purchase this adhesive from Amazon; I have not found any pharmacy that carries it.

Torbot Skin Tac Wipes (Adhesive barrier wipes)

Freestyle recommends the use of a skin barrier wipe to assist the Libre 2 adhesion in difficult cases.



- These wipes contain a sticky polymer dissolved in alcohol.
- Use a wipe the application site. Avoid getting it in the cat's fur.
- Let the skin dry until it is sticky to the touch, then apply the Libre sensor.
- Use a makeup remover pad soaked with alcohol to clean up any excess in the fur and on your fingers.
- You can purchase these wipes from Amazon if you can't find them in a pharmacy.

Apply the sensor

The first time you apply a sensor to your cat is the hardest. I suggest starting with a helper to hold and calm that cat. As you and the cat get more experience, you will find it easier to apply the sensor without help. For the first sensor, you should have the vet assist you.



- Freestyle's self help guide is here: [freestyle-libre-sensor-application-guide.pdf](https://freestylelibre.com/~/media/Files/2017/08/freestyle-libre-sensor-application-guide.pdf)
- Peel back the top of the sensor container, unscrew the applicator from its plastic cap.
- Align the slot in the applicator with the ridge in the sensor container and press down to load the sensor in the applicator and pull it out of the container.
- Apply Vetbond if you desire (I recommend it.)
- Press the applicator down in the application site firmly, you will hear a click. This is usually when the cat will react.
- Hold the sensor down in place for 30 seconds. Peel the adhesive fabric away from the applicator using a fingernail, forceps or some other thin, flat tool.
- Scan the sensor with your phone app to activate it and wait for 60 minutes for the app to start giving readings.

Blood from the sensor

Sometimes you will see some blood on the top of the sensor. You may have hit a capillary (best case) or a vein (worst case), the difference will be the amount of blood and how long the bleeding persists.

- Minor bleeding is possible as the sensor's small needle is inserted under the skin.
- To minimize bleeding, do not push down on the applicator until it is correctly positioned over the prepared site.
- If bleeding is minor and stops, the cat can continue to wear the sensor.
- If bleeding is excessive, or does not stop, remove the sensor immediately and apply a new sensor at a different site.
- Contact your vet if the bleeding is persistent or causes concern.

Did it work?

Once the sensor has started sending readings, it will become clear if there was a problem with the attachment of the sensor.

- The best indicator is a large number of unexplained “sensor error” events (glucose has been fairly stable and has not been too high or low) in the first few hours after attachment.
- Incorrect glucose measurements can also be an indication, but can only be confirmed if you also have a blood glucose monitor.
- You may also quickly get a “replace sensor” message in the Libre app.
- This is why it is always useful to have two spare sensors, giving a backup if the first one has issues.

Maintaining the sensor long term

Unless you are a millionaire, sensors are not cheap. You want to make them last as long as possible. My cat had his first sensor removed in about 2 hours. At its worst, I had to replace 5 sensors in two weeks. Here are a few tips you can use to extend sensor life.

How long can I expect it to last?

Libre 2 sensors are built to last about 14 days. This is an ideal, and very rare case when you use a sensor on a cat. Since the fur regrows under the sensor, slowly dislodging it, you can expect at best 10 to 11 days before you need to replace it.

Cat licking and saliva

The sensors are supposed to be water-proof. Cat saliva is a very different story. Some cats do not mind sensors and leave them alone. My cat is obsessive about grooming and will lick the sensor any time he gets a chance.

- Cat saliva can interfere with the filament, causing incorrect glucose measurements.
- Cat saliva quickly dissolves the adhesive (including Vetbond) causing the sensor to detach prematurely.

Sensor movement

It is really amazing how loose cat skin is and how far it can move. The sensor will move as well.

- If I placed a sensor behind the ribs, on the abdomen, I have seen it move from over the ribs to on top of the back leg.
- If I placed the sensor behind his front leg, I have seen it move to on top of his shoulder.
- In both cases above, this causes stress on the adhesive and will contribute to it coming detached.
- You want the sensor to stay in one place as much as it can, and flat on the body.

Coban Bandage

Coban bandages are used by vets, are elastic and adhesive using a hook and loop mechanism. A couple of wraps of this bandage can cover the sensor and assist in preventing it from moving. They are also absorbent, though not as effective as a gauze pad, which can be used under the bandage.



- They are available from Amazon and are fairly inexpensive.
- [Amazon.ca: Coban bandage](https://www.amazon.ca/dp/B000APR080)
- You will not find these in a pharmacy.
- I use the 3-inch-wide bandage.
- They do tend to bunch up and need replacing every few days.

Surgery recovery shirts

There are recovery shirts for cats that have had abdominal surgery. These can stop the cat from licking the sensor.



- They are available from Amazon and are fairly inexpensive.
- [Amazon.ca: cat abdominal surgery](https://www.amazon.ca/cat-abdominal-surgery)
- Some shirts have an opening at the top of the neck which I found convenient for an injection site.
- I bought four so I could rotate them once a week.
- Be sure to select the correct size for your cat when ordering.



- I found the leg straps that keep the shirt from riding up got in the way when he used the litter box and cut them off.
- This required a Coban bandage to keep the shirt in place and the sensor from getting exposed.
- Unfortunately, he figured out how to roll the bandage up and expose the sensor fairly quickly.

Harness

I moved back to placing the sensor above his shoulder, and a recovery shirt would have a pretty high chance of exposing the sensor. I needed a better option. This halter seems ideal for any sensor placed either behind the leg or on the shoulder.



- I still use Coban bandage to keep the sensor from moving. It is under the halter. The bandage doesn't bunch up as much.
- I removed the handle on the top of the halter.
- You can find these halters on Amazon. [Amazon.ca: cat halter](https://www.amazon.ca/cat-halter)

Removal

I was really concerned about having to remove a sensor. The adhesive on the sensor (with Vetbond, etc.) can be pretty effective. The first sensor my cat removed himself left him with missing skin and I knew that I would have to find a much gentler method.

Removing the sensor, but leaving the adhesive patch (let the cat self remove)

The first couple of times, I tried to remove the sensor as a whole. My cat did not like the tugging on his skin. Then it became clear that it was easier to remove the sensor from the adhesive fabric and deal with the fabric later.

- Hold on to the edge of the fabric and just peel the sensor off it.
- Make sure you check to ensure the filament is still attached to the sensor.
- Now, you just need to remove the adhesive fabric. It's a lot easier to manage.
- If your cat can reach it, let them lick the fabric and dissolve the adhesive.
- If the cat gets too aggressive, it may cause skin irritation.

Manually removing using olive oil

If you have not used Skin Tac, then you can use any oil you would normally use for cooking to work against the adhesive.

- I use a cotton swab, dip it in the oil to soak the swab, and then gently lift the edge of the fabric, rub the swab in the space between the skin and the fabric, slowly peeling the fabric back.
- As you get more space, you can rotate the swab to get oil on the fabric first, and second to slowly apply pressure on the skin to peel it from the fabric.
- Depending on how slowly you do this, it is possible to remove the fabric with a minimum of skin loss or irritation.

Using rubbing alcohol

I have found this method equally effective against Skin Tac, Vetbond and the sensor adhesive, no matter how long the sensor has been attached.

- Soak a makeup remover pad with rubbing alcohol.
- Place the pad on the adhesive fabric and let it sit for a few minutes.
- Remove the pad and peel the fabric off the skin slowly.
- Use more rubbing alcohol on the pad to clean the Skin Tac from the attachment site and surrounding fur.

Other adhesive removers?

There are other adhesive removers you can use, they vary in effectiveness and cost. I have tried them, but found cooking oil or rubbing alcohol to be more effective and less expensive.

- You can find these products on Amazon here: [Amazon.ca : bandage adhesive remover](https://www.amazon.ca/s?ref=nb_sb_nss&url=search-alias%3Fbandage-adhesive-remover)

Following up a difficult removal - Polysporin triple

In cases where you have irritation or missing skin after a sensor removal, you will need to perform a little first aid on the cat.

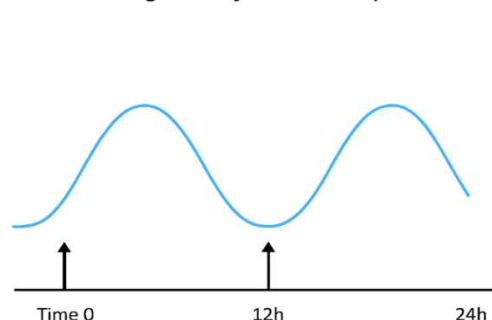


- I use Polysporin Triple.
- You can find it in any pharmacy.
- It works well, if you can keep the cat from licking it.

Understanding curves and what to expect

When I first started, I had an ideal concept of a curve. I thought it should be a smooth sine wave, similar to the image below.

Expected cat glucose curve over 24 hours with feeding and injection every 12 hours



- Reality is much less repeatable. The cat may eat less one day than the next, have less exercise and so on. There are still peaks and troughs, but they are much less regular.
- Every cat is different, but so is every day. You cannot always be perfect, but you can at least get close.
- Remember that insulin can last in the body for up to 36 hours. You need at least 3 successive injections before you reach a point where the dose is working optimally.
- If you change the dose, you should continue for 3-5 days before considering changing it again, unless you get a low glucose (under 4-5 mmol/l).

Here is a sample curve where the cat ate about the same amount and got the same insulin dosage over 3 days.



Leaky liver and bounces

My cat was diagnosed after being diabetic for a long time. At times, it seems that I am fighting his liver, even though the blood work did not indicate any problems with it.

- In diabetic cats the liver becomes relatively resistant to insulin's suppressive effect on glucose production; the liver continues to release glucose into the blood even when peripheral tissues are insulin-treated, so the liver is effectively "leaky" for glucose.
- The liver may release glucose at times when blood glucose is low, or dropping quickly. This will cause a "bounce" or spike in blood glucose.
- After a while of a higher dose of insulin, the liver seems to settle down and I need to reduce the dose to avoid going too low.

Regulation and what it is

Glucose regulation is the coordinated process by which the body maintains blood glucose within a range that supplies tissues with energy while preventing the harms of sustained high or low glucose.

- Daily average glucose levels are lower in well regulated cats; it is a good indicator that regulation is improving.
- Well regulated cats have a higher chance for remission.

Insulin Resistance

The more often and longer the cat's blood glucose level is over 22.2 mmol/L, the more likely the cat will develop insulin resistance, requiring a larger dose of insulin.

- This is why I would rather give a reduced dose of insulin instead of not injecting at all.
- As peak glucose levels get under 22.2 more often, you may find the nadir dropping quickly, requiring a lower dose of insulin.

Cat optimal glucose levels (going for the average versus the nadir)

You always want to adjust the insulin dose so the nadir approaches 6.5 mmol/L, but this may not happen after every injection. It does not make sense to increase the dose too soon because if today is not perfect, tomorrow might be.

- There are other longer-term effects like lower peaks, longer duration of action, lower daily average glucose, all of which are desirable.
- The nadir may not always be perfect, but the average glucose (regulation) over time is something you want to see improvement in.

Be prepared for the unexpected

In any case, if you are dealing with a leaky liver or insulin resistance, the conditions may resolve themselves and you may give an injection that drops the blood glucose too far or too fast. Be prepared for this with a hypoglycemia response kit so you can act quickly.

- As an example, I expect the nadir (6.5 mmol/L) to occur between 9:30 and 10:00.
- If the cat has a glucose reading of 7.0 at 8:00, it is clear to me that I need to feed ½ a can of wet food in gravy and be ready to feed more.
- It's also a sign to me that I need to reduce the insulin dose.

Cost (Budgeting per month, ideal case, Canadian)

- Food (Dry) – 50 (Diabetic)
- Food – Wet – 210 (Tiki Cat + 2 Royal Canin per day @ \$7 per day based on portions, less fussy eaters can be 150)
- Syringes – 30
- Insulin – 85 (Glargine may be cheaper from a pharmacy)
- Bandage – 20 (May not be needed)
- Fortiflora – 36 (Optional)
- Sensors – 330 (assume a sensor lasts 10 days on average)
- Incidentals (Skin Tac, Vetbond, Alcohol wipes, etc.) - 20

Additional Libre reporting using Power BI

One of the weaknesses to the FreeStyle Libre 2 is related to reporting and analyzing blood glucose levels over time.

- LibreLink will allow a user to scroll through the curve, but you only go back 12 hours.
- LibreLink does not display Insulin dosages or notes.
- It can be very difficult to form a “picture over time” that allows proper analysis of the curve to spot problems or successes.
- I created a Power BI dashboard to enable this.

Downloading and installing Power BI

Power BI Desktop is a free Microsoft product that you can use to import and transform large data. The data can then be used to create visualizations, or charts to present the data.

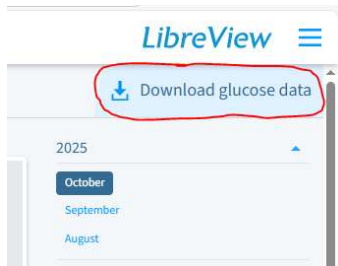
- Here is the product documentation: [Power BI Desktop | Microsoft Power Platform](#)
- Here is the download link: <https://go.microsoft.com/fwlink/?LinkId=2240819&clcid=0x409&culture=en-us&country=us>
- To install Power BI, download the installer and run it.

Downloading the source data from LibreView

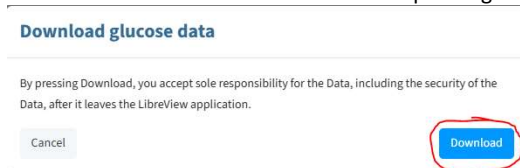
The Power BI Dashboard is designed to import and clean the glucose data from LibreView.

- First, you need to go to the LibreView website: [LibreView](#)
- Complete the login process.

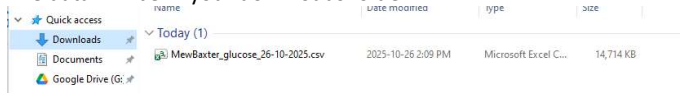
- Click the “Download Glucose Data” link.



- Click the “Download” button and wait. Depending on the amount of data you have, it can take a few minutes.



- The data will be in your downloads folder.



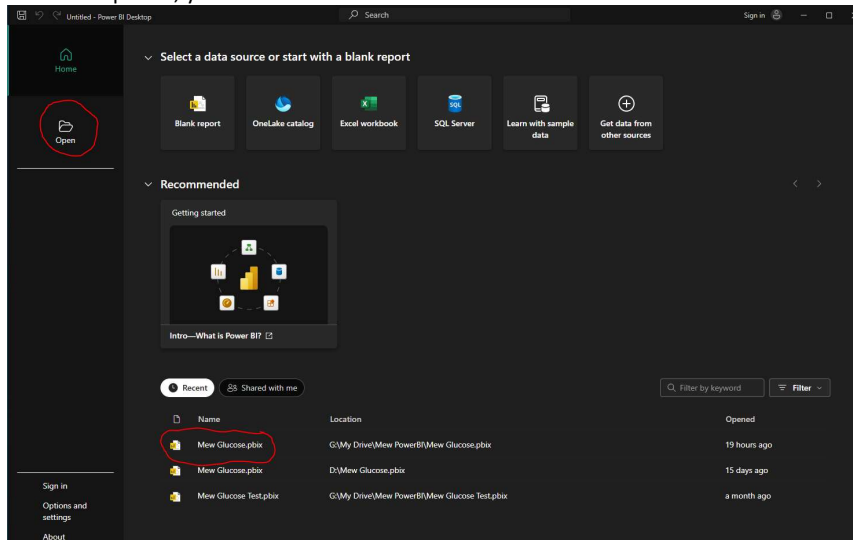
Importing the data to the dashboard

This is a little technical, but I will try to make it easy for you. If you wish to learn more about Power BI, there are plenty of web-based learning resources.

- Run Power BI by locating the desktop icon and double-clicking on it.



- Once running, you need to open the dashboard. The first time, you must use “Open” from the left pane. Once a file has been opened, you can find it in the list of recent files.

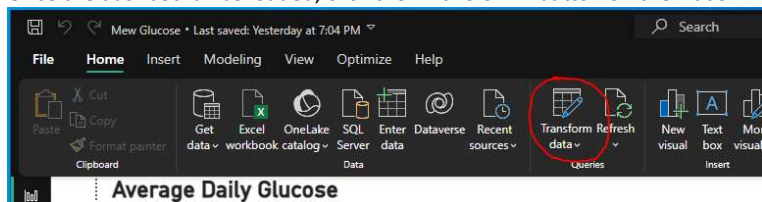


- Hint:** You can rename a copy of an existing dashboard if you are reporting for multiple cats.

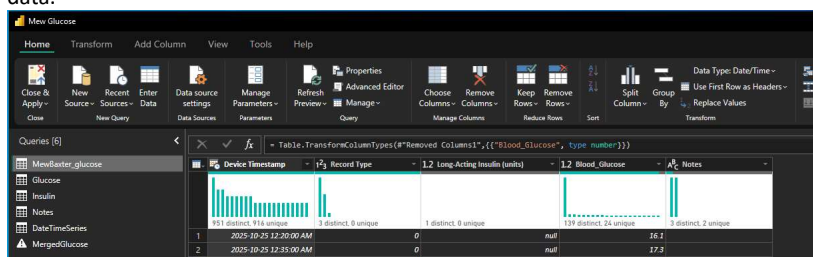
Using the dashboard

Rename the downloaded glucose file to something you can use often and copy it to another folder. You need to make a few configuration changes to the dashboard to make it work for you the first time.

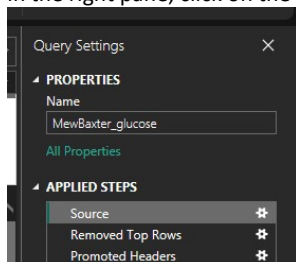
- Once the dashboard has loaded, click the “Transform” button on the ribbon.



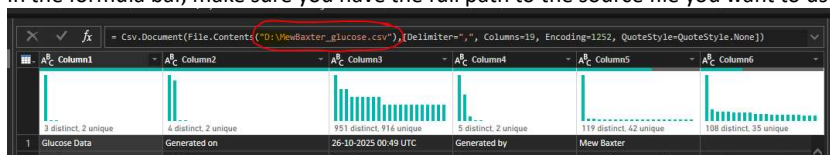
- Click on the “MewBaxter_glucose” query in the left pane and wait for the little graphs to appear above the columns of data.



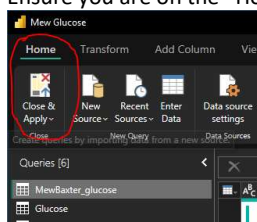
- In the right pane, click on the “Source” step.



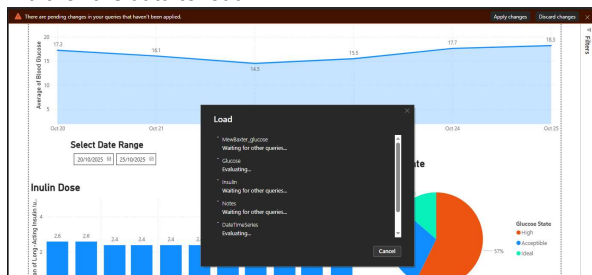
- In the formula bar, make sure you have the full path to the source file you want to use.



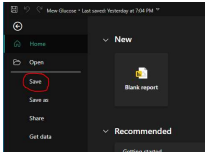
- Ensure you are on the “Home” tab and click the “Close and Apply” ribbon button.



- Wait for the data to load.



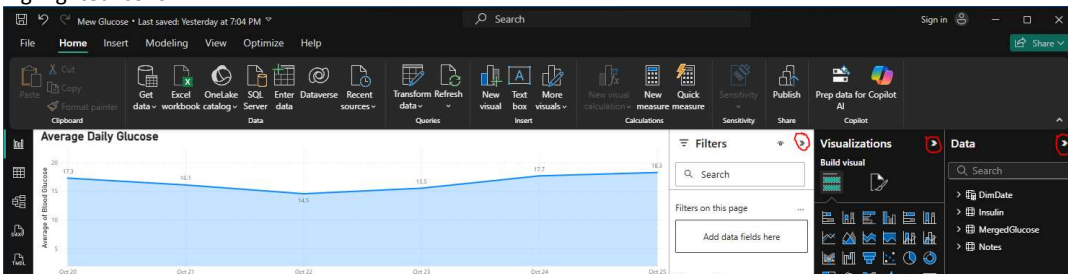
- Once the banner saying “There are pending changes....” has disappeared, choose “File” and “Save” to save your changes.



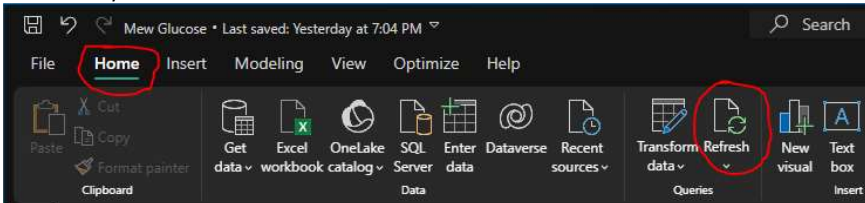
Loading new data into the dashboard

Before you can use the dashboard, you must have:

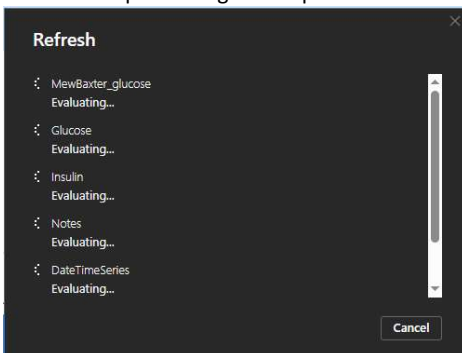
- Properly updated the source file name and path in the dashboard.
- Downloaded, copied and renamed the glucose data from LibreView.
- Run Power BI and open the dashboard you want.
- If any of the three right panes (Filters, Visualization, Data) are opened, you can close them by clicking the little highlighted icons.



- Make sure you have the “Home” tab selected and click the “Refresh” button.



- Wait for the processing to complete.



Individual Reports and uses

Once the data has loaded, you can select the report you want from the bottom bar.

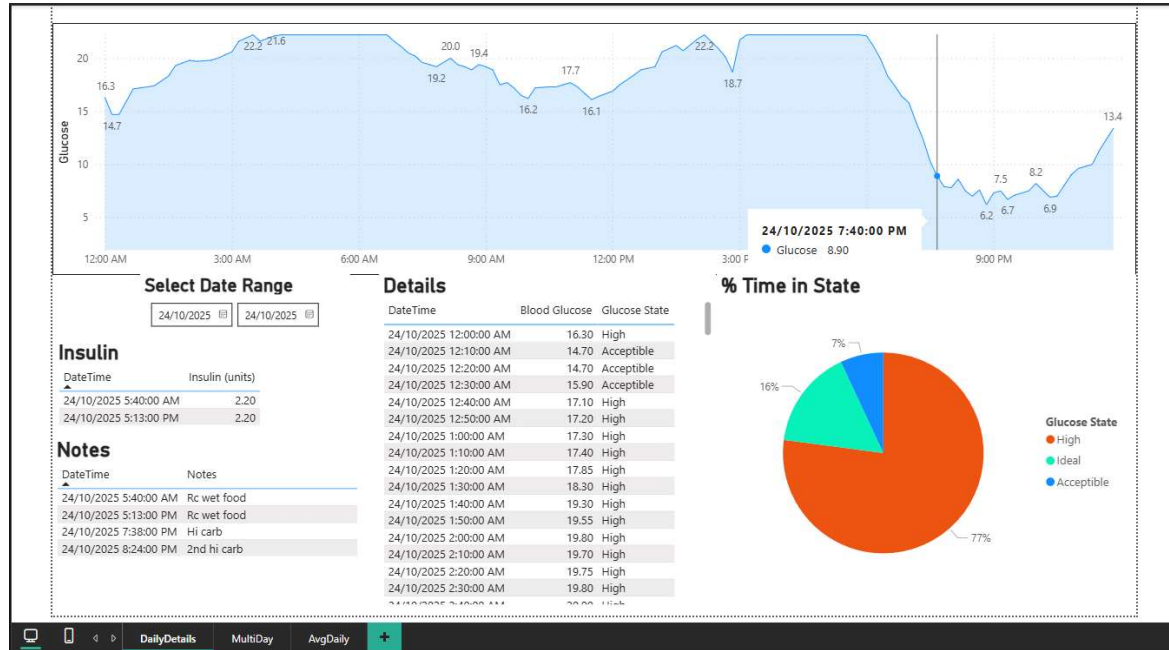


- The process is the same for each report:
 - Select the end, then the start end date.
 - Interpret the data.

DailyDetails

DailyDetails can tell you:

- How much insulin was administered and when.
- Feeding, of which food type and when.
- Blood Glucose readings averaged for 10-minute intervals.
- How much time is spent in each of the glucose ranges:
 - Hypoglycemic – Under 3.8 mmol/L
 - Ideal – 3.8 – 10 mmol/L
 - Acceptable – Over 10 and under or equal to 16 mmol/L
 - High – Over 16 mmol/L
- The owner must record these in the Libre 2 app. There is no automated recording.
 - Insulin is recorded at the time of injection as “Long Term Insulin” with the dose.
 - Feeding and observations are recorded as comments.
- Hovering the mouse over the glucose curve will produce a popout with the blood glucose level at that time, or it can be looked up in the “Details” table by scrolling down.



- In this example, the cat's glucose was at 8.9 mmol/L about 2 hours before the expected nadir, and from the notes, you can see that I fed him some high carbohydrate food at the time.

- In the case of sensor error and a prolonged time without readings, an average value between the last reading and the next one is used and may show as a flat line. Events such as a sensor replacement will show in the chart this way. I like to give the cat a few hours without the sensor if time allows.



- This chart is really only designed to show a single day. Select the date from right to left. You cannot select a day which has no data.

MultiDay

MultiDay can tell you:

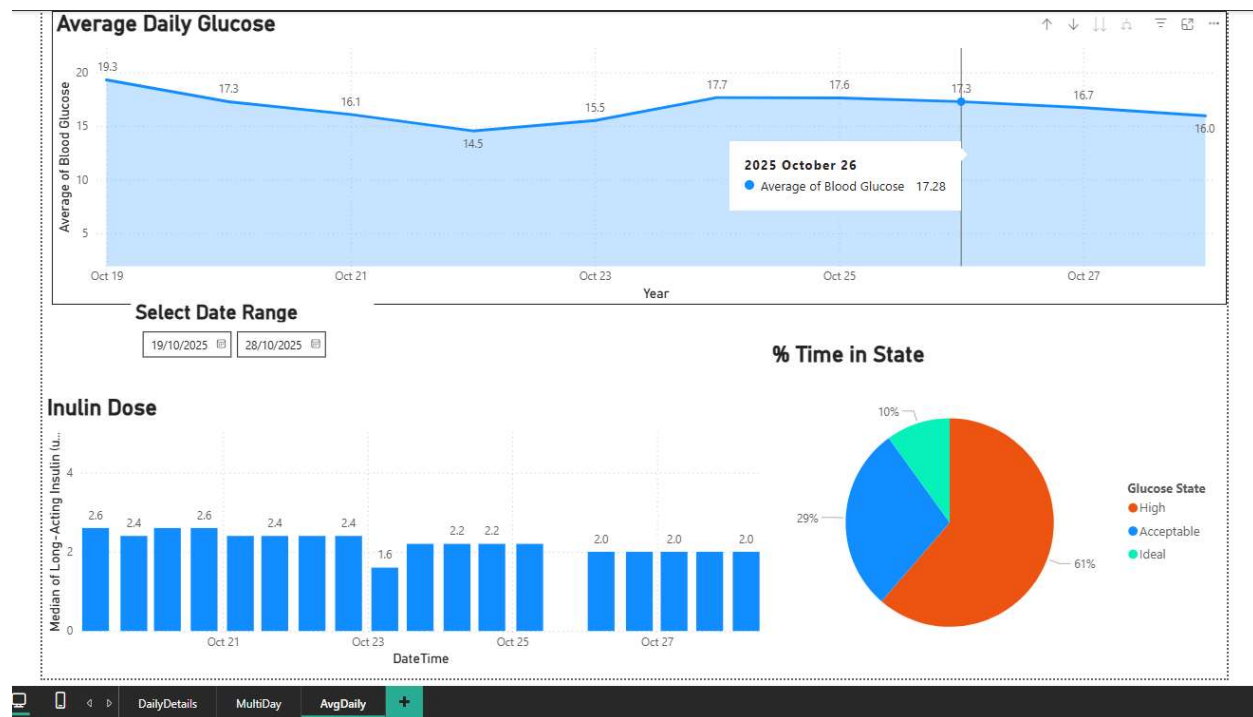
- Nadirs and Peaks during a multi-day period with values listed on the chart.
- Insulin dosages and when administered (if you hover over the bar).
- How much time during that period was spent in each glucose range.
- Of course, the owner needs to keep notes as described in the DailyDetails section.



- Moving the mouse in the glucose graph will show you time and levels for that specific point.
- Note that I also changed his sensor between 9:10am and 3:10pm on the 19th of October.
- Since his peak at 5:00pm on the 19th was lower than expected, I gave him a reduced dose of insulin to be on the safe side.
- There is also a slider on the left so you can change the minimum and maximum glucose values shown on the graph.

AvgDaily

Average blood glucose levels lowering over time give an indication of improving regulation. This report shows average daily blood glucose levels over time.



References, where to learn from

Online information sources

1. The Diabetic Cat - more than just insulin injections. <https://www.youtube.com/watch?v=PcrFF3bZ-Lc>
2. Feline diabetes message board. <https://felinediabetes.com/FDMB/>
3. (2018 AAHA Diabetes Management Guidelines for Dogs and Cats. <https://www.aaha.org/resources/2018-aaha-diabetes-management-guideline-for-dogs-and-cats/cats/>
4. The Veterinary Endocrinologist. <https://www.youtube.com/@theveterinaryendocrinologist>
5. Frequency of diabetic remission, predictors of remission and survival in cats using a low-cost, moderate-intensity, home-monitoring protocol and twice-daily glargine. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11103314/>
6. Purdue Veterinary Medicine: How to Monitor Insulin Treatment for your Diabetic Dog or Cat. <https://www.youtube.com/watch?v=r6B9NUwsmaA>

Feline Diabetes Research Articles:

1. An Individual Approach to Feline Diabetes Care. <https://actavetscand.biomedcentral.com/articles/10.1186/s13028-016-0245-0>
2. Managing Feline Diabetes: Current Perspectives. <https://www.tandfonline.com/doi/pdf/10.2147/VMRR.S125619>
3. ISFM Consensus Guidelines on the Practical Management of Diabetes Mellitus in Cats. <https://journals.sagepub.com/doi/pdf/10.1177/1098612x15571880>
4. Feline Diabetes Update: Insights in Diagnosis and Management. <https://www.canadianveterinarians.net/media/oldhl3cp/feline-diabetes-management-february-2023.pdf>
5. Evaluation of a Quality-of-Life Tool for Cats with Diabetes Mellitus. <https://www.catsanddiabetes.com/articles/academic-papers/>
6. Insulin Glargine and High Protein Diets Linked to Remission. <https://www.catsanddiabetes.com/articles/academic-papers/>
7. Comparison of Diets in Feline Diabetes Management. <https://www.catsanddiabetes.com/articles/academic-papers/>

8. Remission of Diabetes Mellitus in Cats with Ketoacidosis. <https://www.catsanddiabetes.com/articles/academic-papers/>
9. Remission Cannot Be Predicted by Arginine Stimulation Test. <https://www.catsanddiabetes.com/articles/academic-papers/>
10. Canine and Feline Diabetes Mellitus: Nature or Nurture? <https://www.catsanddiabetes.com/articles/academic-papers/>